

Does the Assessment Require the Skill?

Anonymous Author(s)

Abstract

Assessment items carry tags, standard codes, cognitive domains, or rubric criteria, and accountability, procurement, and research read a correct answer as evidence of the tagged competency, which assumes passing requires it. We test that assumption without student data by searching for assessment bypasses: solvers that lack the tagged competency by an explicit restriction on operations or information, never see the key, and pass under the published scoring rule. A bypass shows passing does not require what was denied; it says nothing about students, and finding none does not certify a tag.

We instantiate the search on public-release mathematics items, where tags, keys, and scoring rules are public, under a pre-registered bar. On New York Regents Algebra I and II, language models of two families recover the key well above chance with every given quantity withheld, and the recovery survives a form-preserving change of every option value. A pass there does not require computing on the givens, so the key does not carry the evidence it is read as; whether it requires the tagged operation is left open. Small programs denied the tagged operation match keys on a few cells. Trends in International Mathematics and Science Study (TIMSS) cognitive-domain cells yield no bypass at public sizes. Rates are claimed on items re-read from the printed page, which released text layers often misstate. This is a method demonstration on one subject; the search needs only public tags, item text, keys, and an executable scoring rule.

Keywords: assessment validity, assessment bypass, standards alignment, test-wiseness, partial-input baselines

1 Introduction

An assessment item carries a tag that names the competency it is said to measure: a standard code, a cognitive domain, or a rubric criterion. A correct answer is then read as evidence of that competency. Accountability systems report proficiency by standard, districts buy materials described as standards-aligned and take the aligned assessment as evidence that the materials teach the standard, and research uses a tagged assessment as the outcome measure of an intervention; each use assumes that passing requires the competency. The testing standards treat that reading as a claim to be validated for the intended use [American Educational Research Association et al. 2014], and validity theory treats it as an argument that can fail even when the tag is sincere [Kane 1992, 2013; Messick 1995]: the item may be answerable from a surface cue, from the options without the stem, or from a routine that is never bound to the situation the tag names.

We test the assumption directly and without student data by searching for *assessment bypasses*: solvers that lack the tagged competency by an explicit restriction on the operations they may perform or the information they receive, never see the published key, and still pass under the published scoring rule, so that a pass on those items does not require the part of the competency the solver was denied. Neither restriction kind belongs to a subject: a lexical policy that answers a causal-inference item by locating the phrase random assignment lacks the operation of causal identification, and a reader given a reading-comprehension question and its options with the passage withheld lacks the information the item is about; partial-input baselines [Kaushik and Lipton 2018] already show that such readers beat chance. The claim is one-sided: a bypass says nothing about what students learned, and a search that finds none does not certify the tag beyond the solvers that were tried. Alignment tools, which match item text to a standard, ask whether the item mentions the topic or offers an opportunity to use the competency; psychometric methods ask whether performance discriminates students who have the competency from those who do not, and need student responses. Whether passing requires the competency sits between those questions, needs no students, and is what a bypass refutes.

We run the search first on public-release mathematics items, because several authorities publish standards, keys, scoring rules, and item text in machine-checkable form, and the catalog of shortcut strategies for those items is developed enough to write capability restrictions as small programs. We find bypasses. On New York Regents Algebra I and Algebra II, a language model shown the question with every given quantity replaced by a placeholder recovers the published key well above chance and above the most common key letter, in two model families, and the recovery survives replacing every option value with a different value of the same form. On those items a pass does not require computing on the given quantities; whether it requires the tagged operation is a question the item codings leave open. Small programs that lack the tagged operation match keys on a few cells: a rule that picks the second-smallest numeric option on two cells, and substitution of the options into the equation read from the page image on a small subset of Algebra I. The cognitive-domain cells of the Trends in International Mathematics and Science Study (TIMSS) [International Association for the Evaluation of Educational Achievement 2009], at the 50 to 80 items per domain that are public, yield no bypass. Every claimed rate is measured on items re-read from the printed page, because the text layer of a released PDF often differs from the print, and the language-model rates only

where a second transcription of the page agrees with the first.

Our contributions are the object and the bar, which are general, and the census and its findings, which are the mathematics instantiation.

- **The assessment bypass.** A one-sided, student-free test of any tag: a solver that lacks the tagged competency by an explicit capability or information restriction and passes under the published scoring rule refutes the claim that passing requires what it was denied. Validity theory states the argument, and cognitive diagnosis and test-wiseness research estimate students or list cues; none scores this object (Section 2).
- **A pre-registered bar.** A bypass must fire on at least 10 items with a lower 95% confidence bound above chance, programs are hashed before scoring, the three solver families are Holm-corrected separately, and every claimed row is measured on the item as printed (Section 3).
- **A census.** The first instantiation of the method: 2,405 keyed public-release mathematics items from nine authorities, each tag quoted from its authority, scored by item-writing cues, test-taking strategies, option substitution into the printed equation, and language models denied the given quantities.
- **Findings.** Language-model bypasses on Algebra I and Algebra II that survive a change of every option value, two catalog rows awaiting replication, one strategy row that failed a powered replication, and a TIMSS null at public n (Section 4).

2 Related Work

Standards labeling matches item text to a code. MathFish and related classifiers ask whether a labeler agrees with a publisher [Lucy et al. 2024]; agreement between labelers is not a test of whether the tagged operation is required to pass. We keep the publisher’s tag and search for programs that pass without it.

Shortcut learning and partial-input baselines [Geirhos et al. 2020; Gururangan et al. 2018; Kaushik and Lipton 2018; McCoy et al. 2019; Poliak et al. 2018] show that models and rules can answer from the wrong slice of the input, and visual question answering (VQA) models can keep the same answer across images [Agrawal et al. 2016]. Choices-only language models already beat majority on several multiple-choice benchmarks, by surface artifact or by abductive reconstruction of the question [Balepur et al. 2024], and later work separates ranking artifacts from question inference and surveys how multiple-choice evaluation is scored [Balepur and Rudinger 2024; Balepur et al. 2025]; our masked-stem condition is that construction on tagged mathematics items, with the question type shown and the given quantities withheld. A domain-naive classifier over stem and option features

trained to pick the keys of released science items reaches human-level guessing, read as evidence that the item is a weak measure of the domain [Watson et al. 2018]; that is the closest program prior to our catalog and strategy families, which use a priori hashed rules rather than a fitted classifier. Contamination of released items in training data is a rival explanation of partial-input hits [Magar and Schwartz 2022; Sainz et al. 2023], and substituting the numerals of grade-school word problems lowers language-model accuracy [Mirzadeh et al. 2025]; our option-numeral control is that substitution applied to the option strings. Item-writing and test-wiseness catalogs [Haladyna et al. 2002; Haladyna and Rodriguez 2013; Millman et al. 1965] list surface cues that can determine a choice without content, including using the given numbers without binding them, and have been compiled into item-writing-flaw detectors that flag flaws rather than scoring pass rates [Moore et al. 2024]; some option sets carry information about the key with the stem withheld [White and Zammarelli 1981]; test makers and takers hide and seek keys in middle letter positions [Attali and Bar-Hillel 2003], a position cue and not the numeric rank our lower-central rule picks; plug-in substitution of numbered options is a documented SAT mathematics strategy [Katz et al. 2000], which our image channel runs as a program on the displayed equation.

Q-matrix and cognitive-diagnosis methods estimate which skills an item requires from student response data [de la Torre and Chiu 2016; Tatsuoka 1983], and expert attribute tags on TIMSS 2011 items have been treated as misspecifiable Q-matrix entries under the generalized deterministic inputs, noisy “and” gate (G-DINA) model [Terzi and Sen 2019]. Differential item functioning (DIF) uses examinee groups to detect items that do not measure the same construct across populations [Holland and Thayer 1986]. Language models have been scored on the released TIMSS 2011 grade 8 mathematics items with item response theory against human populations [Fang et al. 2025]; that study asks how well models solve the items, and our TIMSS language-model null should be read next to it. This census uses no response vector, and comparisons to percent correct use only the statistics TIMSS published with the items.

Each of those lines studies one piece of what a tagged correct answer can be taken to mean. Argument-based validity treats the tagged pass as a claim to be supported for its use [American Educational Research Association et al. 2014; Kane 1992, 2013; Messick 1995]; cognitive diagnosis and DIF estimate attributes or bias from student responses; test-wiseness catalogs list cues; choices-only and contamination papers measure models on mostly untagged general-knowledge items; domain-naive guessers, flaw detectors, psychometric benchmarking of models on TIMSS items, and numeral-substitution tests each occupy one vertex of that space [Fang et al. 2025; Mirzadeh et al. 2025; Moore et al. 2024; Watson et al. 2018]. None scores an assessment bypass,

a solver denied part of the tagged competency by construction and run on tagged items under the published scoring rule. The object belongs to tagged assessments in general; this paper adds it, with public-release mathematics items as its first instantiation.

3 Method

3.1 The object

A cell is an authority-by-tag group of items with published keys, and a solver passes an item when it returns the key under the published scoring rule. An *assessment bypass* is a solver policy that lacks the tagged competency by an explicit restriction written into its specification, never sees the key, and passes anyway. What the policy lacks is a fact about its specification, not its internals; the restriction is of one of two kinds, neither tied to a subject. A *capability restriction* removes an operation from the solver's instruction set, so a bypass by such a solver refutes the claim that the pass requires that operation. An *information restriction* removes an input: the solver keeps its operations and is shown the item with part of its content withheld, so a bypass by it refutes the claim that the pass requires operating on what was withheld. It does not refute the claim that the pass requires the relation the tag names, which the solver may apply to what it still sees. Score use rests on the first claim: a correct answer is read as evidence that the student carried the tagged operation out on the content in front of them, and a key that can be matched without that content does not carry that evidence. A family that finds no bypass on a cell leaves the part of the competency it was denied unrefuted there, and a family's pass rate on a cell is a vulnerability measure relative to that documented family, never a probability that a student lacks the competency; the one-sided claim covers only the items a claimed program matches.

3.2 Items and tags

This census instantiates the method on public-release mathematics items. Targets are released summative items and end-of-unit assessments with scoring rules; practice problems are never targets. The census holds 2,405 keyed public items (2,245 selected-response, 160 numeric): New York Regents 1,448; State of Texas Assessments of Academic Readiness (STAAR) 445; Ontario's Education Quality and Accountability Office (EQAO) 149; TIMSS 2011 129; earlier TIMSS public-release 106; Programme for International Student Assessment (PISA) 70; New York State Education Department (NYSED) 27; National Assessment Program Literacy and Numeracy (NAPLAN) 21; Massachusetts Comprehensive Assessment System (MCAS) 10. Each tag is quoted from its authority: Utah Core grades 6–8 use Utah State Board of Education Core Guides, Common Core State Standards (CCSS) seed codes are checked against the same wording, and TIMSS uses the 2011 framework definitions of knowing, applying,

and reasoning [International Association for the Evaluation of Educational Achievement 2009]. TIMSS items keep those tags; we do not assign CCSS codes to them. TIMSS 2011 is the last cycle whose item texts, keys, scoring guides, domain tags, and percent correct are public together with no permission process; from 2015 the International Association for the Evaluation of Educational Achievement (IEA) keeps most items secured, and restricted-use grants cap example items and forbid the comparisons this census needs, so later cycles are not used and recency of showable items comes from released state and provincial tests. Operational Utah RISE items are secure and are not used. A cell enters scoring when it has at least 10 selected-response items.

3.3 Three families of solvers

A family fixes the information and the instruction set its programs receive, and therefore the restriction each carries: the catalog rules, the strategy programs, and the image-channel substitution lack the tagged operation because it is not in their instruction set, and the partial-input language models keep the operations and are denied the given quantities. Each family was fixed before its pass rates were computed: the catalog family is the (rule, cell) list locked in the protocol draft, the strategy programs were hashed before scoring and later parsing edits did not overwrite version-1 rates, and the language-model family was scored the day it was registered. Hashes, timestamps, model revisions, seeds, and filter counts are in Appendix A.

Catalog rules (*capability-restricted*). Fifteen a priori programs taken from the item-writing and test-wisness catalogs, each with a citation and no fitted parameter, scored on every item they apply to: option length, position, absolute terms, a signed variant of the catalog numeric-central-value cue, stem–option overlap, stem-number copying, and unbound formulas (percent of, unit rate, $y = kx$, and the rest of the catalog) applied to stem numerals without binding a symbol to the situation. The catalogs state the central-value cue as: among numeric options, the key is often a middle value [Haladyna et al. 2002; Millman et al. 1965]. The variant we score, *lower-central*, parses a leading signed number from each option and, if at least three parse, sorts them and picks the option at index $\lfloor (n - 1)/2 \rfloor$, the second-smallest of four. The catalog median of four numbers is the average of the two central values, which is not an option, so a discrete program must pick one side; that index is fixed in the program and not fit on these pass rates, and a protocol draft records option numeric rank as a searched feature rather than this exact index. The same index is the tie-break inside the strategy programs. A second class of catalog program is fitted or searched: a library of surface-cue programs enumerated to bounded complexity plus greedy and evolutionary search (1,847 programs, Appendix A.3), fit on state items and scored

on TIMSS or the reverse, with fourteen programs selected on the training corpus and held-out accuracy only.

Adaptive strategy programs (capability-restricted).

Five programs written from the same catalogs. A distractor-derivation hub picks the option from which the most other options can be derived under a fixed set of error transforms. Constraint elimination removes options that a shallow stem keyword rules out (a negative count, a probability outside $[0, 1]$, a mean outside the listed data). Arithmetic closure applies $+$, $-$, $\times$, $\div$, squares, and square roots to the stem numbers to depth two and picks the option that first appears. Backsolving parses a one- or two-variable equation from the stem and picks the unique option that satisfies it. A fixed-order portfolio runs those four. None binds a symbol to the situation the standard names. Text-layer backsolving enters the family only when a cell has at least 10 unique fires, which none does, so a later addendum, hashed before scoring and outside the family, has a vision-language model (VLM) transcribe the displayed equation from the page image and substitutes each option symbolically; the model is not asked to solve, and an invented formula or an extract that does not match the page is an abstention. Tie-breaks, the transform set, and the frozen hashes are in Appendix A.1.

Partial-input language models. Here the restriction is on information. A local instruction-tuned language model sees a slice of the item and picks the option letter with the highest next-token log-probability; there is no chain of thought and no free generation. Two model families, Qwen2.5 (7B-Instruct and 14B-Instruct) and Phi-4, are capable scorers; Mistral-7B-Instruct-v0.3 is a weak-solver check. Two prompts are scored: options only, with the stem withheld entirely, and *masked stem*, in which every numeral, number word, fraction, percentage, decimal, and value binding in the stem is replaced by the placeholder $[N]$ while question wording, units, operators, and the option strings are kept. Options only is an information-ladder rung; the masked stem is the bypass. A mechanical clean-option filter first drops items whose options carry exam footers, item codes, country-table fragments, options longer than 120 characters, or bleed from the next stem, keeping 1,496 of 2,245 selected-response items, and items on which the masker withheld nothing are excluded because the tagged operation may still be performable, leaving a primary population of 1,215 items with at least one withheld quantity. Model revisions, the letter-token rule, the filter, and the masking hashes are in Appendix A.2.

Two kinds of probe attach to this family, and neither is a family test. Contamination probes are the mean token log-probability of the original unmasked stem plus options, administration-year buckets, option-letter permutation, and a second masking pass that also hides named figure types, variable bindings, and leftover coefficients. The pre-registered *option-numeral isomorph control* rescores the same masked stems with every numeral in the option strings

replaced by a value-changing, form-preserving isomorphism (same digit count, sign, decimal places, and comma grouping; the same replacement wherever a literal repeats within an item; one arm preserves the rank order of the options' leading numbers and one scrambles it), with the key letter unchanged: a rate that holds is read from the form of the options, and a rate that falls to chance was read from their values (Appendix A.2).

3.4 The witness bar and the Holm families

A program is a *witness* on a cell only if it scores at least 10 items there and the lower 95% bootstrap confidence interval of its pass rate exceeds format chance: one divided by the number of options, or, for a program that eliminates options, the mean of one divided by the options remaining, so 0.25 on four-option items with no elimination. Searched, fitted, hub, and portfolio programs must also reach the 95th percentile of random programs of matched complexity, which we call the machine bar; random-weight programs are the baseline and never witnesses. The language-model bar adds the modal key letter: the lower bound must exceed both chance and the frequency of the most common key letter in that population, on the withheld-quantity items. A singleton match is logged, not claimed.

Multiplicity is controlled inside each family and never across them. The catalog family is every (rule, cell) pair with at least 10 scored items and positive chance, 105 tests. The strategy family is every (program, cell) pair at the same floor, 67 tests. The language-model family is every (model, cell) pair on the withheld-quantity masked-stem population with at least 10 items, 68 tests (four models, 17 cells); options only is not a member. Each family is Holm-corrected at $\alpha = 0.05$ on its own, so a row that clears the per-cell bar in one cell is never billed as a family-wise discovery, and a 172-test pool of the catalog and strategy tests is reported as a sensitivity only. The image-backsolving addendum and the probes above are outside every family; the addendum's bar is at least 10 fires and a lower 95% interval strictly above chance.

3.5 Controls

Before a census null means anything, the same searcher has to find witnesses where they are known to exist. We run a synthetic set with planted item-writing flaws (circular: the planted cues are the features) and, as the primary control, public multiple-choice corpora whose authors already published an options-only or question-ignored number [Lai et al. 2017; Mihaylov et al. 2018; Talmor et al. 2019; Zellers et al. 2018].

3.6 Print fidelity

Every rate was first computed on the PDF text layer of the released form, and a 20-item hand audit against the printed pages found that layer wrong on about half of the sampled

rows (Section 4.1). Two further layers follow, and claims are made on the second.

Filter. For every item in the cells that cleared on the text layer (955 items: the two algebra masked-stem populations, the two lower-central fired sets, and the unclaimed Phi-4 geometry cell), a VLM (Qwen2-VL-7B-Instruct, the model used for image back-solving) transcribes the stem and options from the rendered page image; it is never asked to solve or to pick a letter. Transcription and text layer are normalized identically and compared token by token: the stem must reach token F1, precision, and recall of at least 0.8, every option must match at 0.9, and the option counts must agree, and the item is then *print-faithful*; an unparseable transcription is *unverified* and is excluded from both subsets. Thresholds, prompt, and normalization were frozen and hashed before any population item was scored, after development on 40 items outside the population, and were calibrated on the 20 hand-audited items: 17 of 20 agree on the binary verdict, above the pre-registered floor of 16, so the thresholds were not adjusted. Every row that cleared on the text layer was recomputed on the faithful subset and on its complement from the saved per-item flags and predictions, with the same bootstraps and bars. The faithful subset is not a random sample of the cell: it over-represents administrations before 2023, whose PDFs do not encode operators as digits, and items with word options, and a transcription error moves an item out of it and never in (Appendix A.4).

Repair. The filter keeps a selected subset; the repaired layer replaces the text layer for every item in the two algebra masked-stem populations (606 items). The transcription above becomes the item’s stem and options, and a second transcription of the same page by the same model, with a reworded prompt and a larger crop, is compared to the first under the thresholds above. An item on which the two agree is *repaired-verified*; every other item is *repaired-unverified*. The key letter is always the census key. Every masked-stem, item-class, and isomorph row is then rescored on the repaired stems and options with the same models, seed, and letter rule. The procedure, prompt, thresholds, and the amendment that re-runs Qwen 14B on the isomorph arms were hashed before the second transcription and before any scoring on this layer (Appendix A.6). As a stricter verification, pre-registered after the repaired layer was scored, a third transcription of every page by a model of a different family, LLaVA-NeXT Vicuna 13B, is compared with the first Qwen transcription under the same thresholds, and an item on which the two families agree is *cross-family verified* (Appendix A.6). Language-model claims are made on the repaired-verified population; the cross-family verified population is a stricter sensitivity, the full repaired population a looser one, and the text-layer and filtered rows are an audit trail (Appendix B). The two catalog rows that cleared on the text layer are rerun on the

stored first transcription of their options, without a second transcription.

3.7 Two codings of what an item asks

The language-model rows carry two codings of item demand. The first is the publisher’s tag: the NYSED rating guide of every Regents administration maps each question to a CCSS cluster, such as A-REI.B, so every item in the two algebra populations has one. Under a pre-registered rule, the verbs of each cluster heading classify the cluster as *execution* (solve, rewrite, factor, complete the square, create equations, compute, graph, perform arithmetic operations, and the like: operating on given quantities), *recognition or interpretation* (identify, interpret, recognize, compare, explain, understand), or *mixed* when the heading carries both; the rule and the classification of all 33 clusters were hashed before any prediction was joined to a tag (Appendix A.5). That is a classification of the tag’s text, not of the item. The second coding is item-level: two language-model coders, GLM 5.2 and Kimi K3, each read every Algebra I item’s stem and options, with the key, every prediction, and the cluster split withheld, and assigned one of the same three codes under one fixed rubric (Appendix A.7); agreement between the coders and with the tag is reported as Cohen’s κ . The masked-stem rows are recomputed on each tag class, on the items both coders give the same code, and on the intersection of those with the tag, at the unchanged bar. A statement about the tagged operation needs the items both coders mark as requiring operation on the givens; a statement about withheld information needs no such population.

4 Results

4.1 The text layer against the printed page

A 20-item hand audit, 10 geometry lower-central hits and 10 Algebra I masked-stem hits, compared each census row to the source exam PDF by eye. Nine rows match stem, options, and key. Eight have footer, figure-label, or previous-item glue on an option or stem. Two are truncated. One Algebra I row has the printed stem and a different option set (June 2023 item 3: printed lines $y = 2x - 2$ and three neighbors; census inequalities). No sampled PDF was missing.

The mechanical filter of Section 3.6 then classified all 955 items in the cells that had cleared on the text layer: 434 print-faithful, 502 not faithful, 19 unverified. The text layer is wrong in two main ways: glue from footers, diagram labels, table cells, or the next item (241 items), and option sets or stems that do not match the printed item (230), most of them Regents forms from 2023 onward whose PDFs encode =, +, and − as the digits 5, 1, and 2. The faithful fraction is 0.52 on Algebra I, 0.48 on Algebra II, 0.36 on the geometry lower-central set, and 0.37 on EQAO grade 6. Recomputed on the faithful subset (Table 12), Algebra I masked-stem cleared on all three models at $n = 184$, Algebra II cleared chance but

not the modal-letter bar at $n = 115$, geometry lower-central kept its rate and lost its interval at $n = 99$, and EQAO grade 6 cleared at $n = 14$. In Algebra I, 40 percent of faithful items have word options against 10 percent of not-faithful items, so the filter selects items as well as removing dirt.

The repaired layer removes that selection for the two algebra populations. Of 606 items, 431 are repaired-verified and 175 are not (Algebra I 263 and 95; Algebra II 168 and 80). Of the 431, 246 had been print-faithful and 185 had not: the repair recovers 185 items the filter dropped, and 53 items the filter kept do not verify. Fifteen items have no usable transcription and 17 more have no withheld quantity in the transcribed stem, so the withheld-quantity repaired population is 574 items, 343 Algebra I and 231 Algebra II, with 261 and 167 verified, verified fractions of 0.73 and 0.68 against the filter’s 0.52 and 0.48. The cross-family check of Section 3.6 is stricter: the LLaVA-NeXT transcription agrees with the first Qwen transcription on 160 of the 606 items (Algebra I 102, Algebra II 58), 153 of them already repaired-verified and 7 not, and 5 two-page crops exceeded the LLaVA context and do not verify. With a withheld quantity, the cross-family verified population is 101 Algebra I and 57 Algebra II items. The first transcription also replaces the options of the two catalog rows where it is stored (277 of 279 geometry and 38 of 40 EQAO grade 6 fired items).

4.2 Language models denied the given quantities

Masked stem plus options. The withheld-quantity population on the text layer is 1,215 of 1,487 clean scored items, of which Algebra I contributes 358 and Algebra II 248; on the repaired layer 574 of the 606 algebra items keep a usable transcription and a withheld quantity, 261 and 167 of them verified. Table 1 gives the rows on all three layers. On the repaired-verified population Algebra I is $141/261 = 0.540$ [0.479, 0.598] on Qwen 7B, $144/261 = 0.552$ [0.494, 0.613] on Qwen 14B, and $129/261 = 0.494$ [0.437, 0.559] on Phi-4, against chance 0.250 and modal B 0.287. Algebra II is $75/167 = 0.449$ [0.371, 0.527], $72/167 = 0.431$ [0.353, 0.503], and $69/167 = 0.413$ [0.335, 0.491], against chance 0.250 and modal C 0.275. All six rows clear both bars, and all six clear again on repaired-all (Algebra I $n = 343$, modal C 0.271; Algebra II $n = 231$, modal C 0.286). On the cross-family verified population, the strictest, Algebra I clears on all three scorers ($57/101 = 0.564$ [0.455, 0.653], $57/101 = 0.564$ [0.465, 0.663], and $54/101 = 0.535$ [0.426, 0.634] against modal C 0.297), and Algebra II clears on Qwen 7B ($26/57 = 0.456$ [0.316, 0.579]) and Qwen 14B ($25/57 = 0.439$ [0.316, 0.561]) against modal C 0.281 but not on Phi-4, whose $21/57 = 0.368$ [0.246, 0.491] clears chance and not the modal letter. The Algebra II Phi-4 row is therefore not robust to the stricter verification at $n = 57$; the three Algebra I rows and the two Algebra II Qwen rows are. One-sided binomial p against chance is below 10^{-16} for every Algebra I row and below 10^{-5} for every Algebra II

row; the pre-registered bar is the bootstrap interval, and those p values are a description.

The layers matter to the rows. The repaired Algebra I rates (0.540, 0.552, 0.494) are above the text-layer rates (0.388, 0.416, 0.422) on every model: the dirty layer deflated this row. Algebra II had cleared on the text layer (0.367, 0.359, 0.347 at $n = 248$) and failed the modal-letter bar on its 115 print-faithful items, where the point estimates (0.348, 0.383, 0.348) stayed close to the text-layer rates but the lower bounds (0.261, 0.287, 0.270) fell below that subset’s modal frequency 0.296; the filter had removed items, not the effect, and the repaired layer, at 167 verified items, clears the bar on every model.

Holm at $\alpha = 0.05$ in the 68-test language-model family, against chance on the text layer, rejects both Qwen Algebra I rows, both Qwen Algebra II rows, Phi-4 Algebra I, Phi-4 Algebra II, Phi-4 geometry, and Qwen 7B TEKS grade 8; the repaired layer is a rescoring of two of those cells on a corrected text and adds no test. Geometry on Phi-4 meets the text-layer cell bar ($67/166 = 0.404$) but not the split on items whose key is not the cell’s modal letter (0.229, $n = 118$, CI [0.161, 0.305]) and is not claimed; Qwen 7B clears Texas Essential Knowledge and Skills (TEKS) grade 5 and grade 8 on that scorer only; Qwen 14B TIMSS knowing rejects Holm against chance and does not beat its population modal 0.409. None of those cells was re-read from the page, and none is claimed (Table 9). Scorers are not averaged: the claim is that a reader exists in the Qwen2.5 family and one outside it, Phi-4, a 14-billion-parameter instruct model, and that a weaker reader does not. Mistral-7B-Instruct-v0.3, on the text layer, is 0.302 [0.257, 0.349] on Algebra I, which clears chance but not modal B 0.271, and 0.262 [0.210, 0.319] on Algebra II, which clears neither; its with-stem letter-argmax ceiling matches its masked-stem rate (Algebra I 0.304, Algebra II 0.290; Table 11), so that miss is capability, not a missing cue. One human rater on the pre-registered 40-item Algebra I masked-stem sheet scored $15/40 = 0.375$ (bootstrap 95% CI [0.225, 0.525]), an uninformative null; the claim is a model fact.

Option-numeral isomorphs. The masking leaves the option strings untouched and released Regents forms are widely mirrored, so a memorized item could carry the rate. Under the rank-preserving isomorph control (Section 3.3), with the original and the perturbed letters scored in the same run from the same model load so that no reproduction floor applies, every row clears the bar on the repaired-verified population (Table 2): Algebra I 0.500 [0.438, 0.562], 0.465 [0.404, 0.527], and 0.419 [0.354, 0.477] against modal B 0.288; Algebra II 0.412 [0.339, 0.485], 0.394 [0.321, 0.473], and 0.382 [0.315, 0.461] against modal C 0.273. Four items whose perturbation raised or timed out are excluded (n becomes 260 and 165 verified, 342 and 228 all); items whose options carry no numeral are unchanged and stay in the rate. The paired drop on items with an option numeral is

Table 1. Masked-stem rows (given quantities withheld from the stem, options shown) on the two claimed cells, three scorers, four layers from strictest to loosest. Cross-family verified: the first Qwen2-VL transcription of the page agrees with a LLaVA-NeXT transcription of the same page; the strictest population, a sensitivity. Repaired verified: the first transcription agrees with a second Qwen2-VL transcription; this is the claimed population. Repaired all: every item with a usable page transcription, verified or not. Text layer: the PDF text layer as released. Chance is 0.250. Modal is the modal key-letter frequency of that population (cross-family: C 0.297 Algebra I, C 0.281 Algebra II; verified: B 0.287, C 0.275; all: C 0.271, C 0.286; text layer: B 0.271, C 0.278). Bar: $n \geq 10$ and lower 95% CI above chance and above modal. Every row clears on the repaired verified, repaired all, and text layers; on the cross-family verified population every row clears except Phi-4 Algebra II ($\dagger$), whose lower bound 0.246 is below modal C 0.281. The print-faithful filtered rows, on which Algebra II did not clear, are in Table 12.

Claim	Scorer	Cross-family verified $n = 101 / 57$	Repaired verified (claimed) $n = 261 / 167$	Repaired all $n = 343 / 231$	Text layer $n = 358 / 248$
Algebra I	Qwen 7B	0.564 [0.455, 0.653]	0.540 [0.479, 0.598]	0.504 [0.449, 0.560]	0.388 [0.341, 0.439]
Algebra I	Qwen 14B	0.564 [0.465, 0.663]	0.552 [0.494, 0.613]	0.519 [0.464, 0.574]	0.416 [0.363, 0.469]
Algebra I	Phi-4	0.535 [0.426, 0.634]	0.494 [0.437, 0.559]	0.493 [0.443, 0.545]	0.422 [0.374, 0.472]
Algebra II	Qwen 7B	0.456 [0.316, 0.579]	0.449 [0.371, 0.527]	0.433 [0.368, 0.498]	0.367 [0.306, 0.423]
Algebra II	Qwen 14B	0.439 [0.316, 0.561]	0.431 [0.353, 0.503]	0.416 [0.351, 0.481]	0.359 [0.298, 0.415]
Algebra II	Phi-4	0.368 [0.246, 0.491] [†]	0.413 [0.335, 0.491]	0.407 [0.342, 0.472]	0.347 [0.286, 0.407]

−0.049 [−0.117, 0.022] for Qwen 7B, −0.103 [−0.166, −0.040] for Qwen 14B, and −0.090 [−0.157, −0.013] for Phi-4 on Algebra I ($n = 223$ paired), and includes zero for all three on Algebra II. Every scorer therefore has a form component that survives a change of every option value, on both cells, and Qwen 14B and Phi-4 also have a numeral-specific component of about 0.1 on Algebra I, which could be memorized option values or value cues that survive the masking, such as an option whose magnitude fits the question type; the control does not separate the two. Had the rank-preserving rate fallen to chance, the original rate would not have been a form effect; it did not on any scorer or cell. The rank-preserving arm also clears on every scorer and both cells on the cross-family verified population (Algebra I 0.495, 0.446, and 0.426 at $n = 101$ against modal C 0.297; Algebra II 0.456, 0.544, and 0.439 at $n = 57$ against modal C 0.281), Phi-4 on Algebra II included, whose original rate on that population does not clear. The text-layer control, on which Qwen 14B was excluded by a reproduction floor, is audit trail (Tables 14 and 15).

Contamination probes. On the text layer, 7B masked-stem accuracy by quartile of mean token log-probability of the original unmasked stem plus options is 0.349 in Q1 versus 0.439 in Q4 (Spearman 0.091; 14B Spearman 0.107, Q4 minus Q1 0.120). Recency on all 1,487 masked-stem 7B items is flat: 0.416 / 0.409 / 0.399 / 0.402 across before 2015, 2015 to 2019, 2020 to 2023, and 2024 and later ($n = 178 / 579 / 383 / 331$). Option permutation on the withheld-quantity items (7B) is 0.385 [0.357, 0.412] versus 0.376 unpermuted; Algebra I is 0.405 [0.355, 0.453]. Version-2 masking on the version-1 withheld-quantity overlap is 0.370 versus 0.376, delta −0.007 (McNemar 50 version-2-only correct, 58 version-1-only), so extra masking does not create the channel. The probes and the isomorph control together do not support

memorization of the items as the whole effect; they do not rule out a memorized share.

Which items carry the rows. Table 3 joins the two codings of Section 3.7 to the repaired-verified rows. By the verb class of the publisher’s tag, the execution-tagged Algebra I class ($n = 130$) clears on all three models (0.546, 0.515, 0.477 against modal B 0.323), as does the recognition-tagged class ($n = 105$: 0.514, 0.543, 0.495 against modal B 0.267); on the filtered subset the execution-tagged class had cleared on no model at $n = 75$, and the repaired layer reverses that. On Algebra II the execution-tagged class ($n = 83$) clears on all three and the recognition-tagged class ($n = 57$) on Qwen only (Phi-4 0.368 [0.246, 0.491]). On the cross-family verified population the classes are small: the Algebra I execution-tagged class ($n = 44$) clears on Qwen 7B only and the recognition-tagged class ($n = 43$) on all three, and the Algebra II execution-tagged class ($n = 18$) clears on no scorer and the recognition-tagged class ($n = 28$) on Qwen only.

The item-level coders agree with each other moderately (Cohen’s $\kappa = 0.576$ on 356 items, observed agreement 0.789) and barely with the cluster split ($\kappa = 0.071$ and 0.230; observed agreement 0.461 and 0.564). Both see execution demand in more items than the tag does: 212 and 249 items coded execution against 164 execution-tagged, and of the 151 items tagged recognition or interpretation, 86 are coded execution by each coder and 79 by both, while 18 to 41 of the 164 execution-tagged items are coded recognition. The tag class is therefore not a population of execution items, and we lean on the intersections. On the 142 verified items both coders code execution, Qwen 7B (0.401 [0.324, 0.486]) and Qwen 14B (0.437 [0.359, 0.521]) clear the bar against modal B 0.303; Phi-4 (0.352 [0.275, 0.430]) clears chance and not the modal-letter frequency, and on repaired-all ($n = 188$) clears by 0.011 (lower bound 0.298 against modal C 0.287), which

Table 2. Option-numeral isomorph control on the repaired layer. Every numeral in the option strings is replaced by a value-changing, form-preserving isomorph that keeps the rank order of the options’ leading numbers; the masked stem and the key letter are unchanged, so a rate that holds is read from the form of the options, not their values. Original and rank-preserving letters come from the same model load. Paired Δ is rank-preserving minus original on items with at least one option numeral ($n = 223$ Algebra I, 153 Algebra II, verified). Bar: $n \geq 10$, lower 95% CI above chance 0.250 and above the subset modal-letter frequency (verified B 0.288 and C 0.273; all C 0.272 and C 0.281). Every row clears on both populations.

Claim	Scorer	n	Original	Repaired verified			Bar	n	Repaired all	
				Rank-preserving	Paired Δ [95% CI]				Rank-preserving	Bar
Algebra I	Qwen 7B	260	0.540	0.500 [0.438, 0.562]	-0.049 [-0.117, 0.022]	yes	342	0.471 [0.418, 0.523]	yes	
Algebra I	Qwen 14B	260	0.552	0.465 [0.404, 0.527]	-0.103 [-0.166, -0.040]	yes	342	0.430 [0.377, 0.482]	yes	
Algebra I	Phi-4	260	0.494	0.419 [0.354, 0.477]	-0.090 [-0.157, -0.013]	yes	342	0.409 [0.357, 0.459]	yes	
Algebra II	Qwen 7B	165	0.449	0.412 [0.339, 0.485]	-0.039 [-0.118, 0.046]	yes	228	0.395 [0.329, 0.456]	yes	
Algebra II	Qwen 14B	165	0.431	0.394 [0.321, 0.473]	-0.046 [-0.131, 0.046]	yes	228	0.377 [0.320, 0.439]	yes	
Algebra II	Phi-4	165	0.413	0.382 [0.315, 0.461]	-0.039 [-0.124, 0.039]	yes	228	0.351 [0.289, 0.408]	yes	

is not a margin to claim on. On the 79 items that both coders and the tag mark as execution, only Qwen 7B clears. On the 61 items both coders code recognition or interpretation, all three models clear at 0.738, 0.705, and 0.705 against modal A 0.295, and those rates survive the option-value change (0.639, 0.689, 0.607); the recognition-coded items carry the highest rates on every model.

The statement we make is therefore the one about withheld information: on the items a claimed model matches, a pass does not require the given quantities. The stronger statement, that on items requiring operation on the givens a pass does not require executing that operation, would need the both-coders execution subset to clear on all three scorers on the verified population, and Phi-4 does not clear it; a further coding or a third scorer that cleared it on every scorer would replace the first statement with the second. The reader also uses cues outside the catalog. On the text layer, Qwen 14B Algebra I scores 0.341 when the key is the lower-central option ($n = 82$) against 0.438 when it is not ($n = 276$), 0.454 versus 0.402 for the modal letter ($n = 97 / 261$), 0.370 versus 0.423 for the hub ($n = 46 / 312$), 0.333 versus 0.427 for the unique longest option ($n = 42 / 316$), and 0.442 [0.368, 0.521] on the 163 items with none of the four cues, against chance 0.250; Phi-4 on the same residual is 0.380 [0.307, 0.460]. By option type, Qwen 14B scores algebraic expressions 0.462 ($n = 65$), text 0.452 ($n = 188$), numeric 0.310 ($n = 71$), mixed 0.385 ($n = 26$), and ordered pairs 0.250 ($n = 8$); Phi-4 scores algebraic 0.369, text 0.473, and numeric 0.366 [0.254, 0.479], which clears chance.

An item with the quantities hidden. New York Regents Algebra I, January 2023, item 22 (id nyregents-algebra-i-2023-jan-q22), which NYSED maps to cluster A-APR.B, “understand the relationship between zeros and factors of polynomials”, a recognition-or-interpretation tag; both item-level coders code it recognition or interpretation. Masked stem, with the options as printed on the page:

If the zeros of the function $g(x)$ are $\{[N], [N], [N]\}$, which function could represent $g(x)$?

- A. $g(x) = (x + 3)(x - 4)$
- B. $g(x) = (x - 3)(x + 4)$
- C. $g(x) = x(x + 3)(x - 4)$
- D. $g(x) = x(x - 3)(x + 4)$

Published key: C. On the repaired layer Qwen 7B and Qwen 14B pick C and Phi-4 picks D. The visible cue is the three placeholders in the zero set, which match the three-factor options that include an x : a reading of n factors from n placeholders. The reader may well apply the zeros-and-factors relation to the form of the options; what it cannot do is compute on the given zeros, which it never sees, and a correct answer here is read as evidence that the student matched the printed zeros to factors. In the census text layer of this form, $=$, $+$, and $-$ are non-printing control bytes and the stem’s minus sign is lost; the page transcription restores the operators. The two transcriptions disagree (one drops the $g(x) =$ prefix from each option), so the item is in the repaired-all population and not in the verified one.

4.3 Programs denied the tagged operation

Catalog rules. We ran 350,535 local trials. Fifteen a priori rules on every cell give the 105-test family (one PISA numeric unit-rate row with chance 0 excluded as degenerate); five tests clear the per-cell bar on the text layer, and none survives Holm or Bonferroni at $\alpha = 0.05$ (Bonferroni cut 0.00048). There is no family-wise a priori discovery. Two of the five are lower-central, with one-sided binomial p of 0.0038 (New York Regents geometry, 90/279) and 0.00070 (EQAO grade 6, 20/40) on the text layer. On options re-read from the page image both clear the per-cell bar (Table 4): geometry 80/246 = 0.325, item interval [0.264, 0.382], and EQAO grade 6 18/38 = 0.474, item interval [0.316, 0.632]. On the filtered subset geometry had kept its rate and lost its interval at $n = 99$ and EQAO grade 6 had cleared at $n = 14$; the page-read options restore the n the filter removed. The fired n is the number of items on which at least three option numbers parse: 279 of 408 geometry items and 40 of 53 EQAO grade 6 items on the text layer, 246 and 38 on the page-read options (two items in each cell have no stored

Table 3. Masked-stem rows on the repaired-verified population by what the item asks, three scorers. Cluster tag: the pre-registered verb class of the CCSS cluster heading NYSED assigns to the item (execution on given quantities, or recognition or interpretation of a form). Both coders: the two blind item-level coders agree on that code. Bar: $n \geq 10$, lower 95% CI above chance 0.250 and above the modal key-letter frequency of that subset. The both-coders execution subset clears on Qwen only; the both-coders recognition subset clears on all three and carries the highest rates. Text-layer and filtered rows by cluster class are in Table 13.

Claim	Item class	n	Qwen 7B	Qwen 14B	Phi-4	Modal	Bar
Algebra I	cluster tag: execution	130	0.546 [0.454, 0.631]	0.515 [0.423, 0.600]	0.477 [0.385, 0.562]	B 0.323	all three
Algebra I	cluster tag: recognition or interpretation	105	0.514 [0.429, 0.610]	0.543 [0.448, 0.629]	0.495 [0.400, 0.590]	B 0.267	all three
Algebra I	both coders: execution	142	0.401 [0.324, 0.486]	0.437 [0.359, 0.521]	0.352 [0.275, 0.430]	B 0.303	Qwen only
Algebra I	both coders and tag: execution	79	0.430 [0.329, 0.544]	0.380 [0.278, 0.494]	0.354 [0.266, 0.456]	B 0.304	7B only
Algebra I	both coders: recognition or interpretation	61	0.738 [0.623, 0.836]	0.705 [0.590, 0.820]	0.705 [0.590, 0.820]	A 0.295	all three
Algebra I	both coders and tag: recognition or interpretation	35	0.657 [0.486, 0.800]	0.571 [0.429, 0.714]	0.657 [0.514, 0.800]	D 0.343	all three
Algebra II	cluster tag: execution	83	0.494 [0.386, 0.602]	0.422 [0.313, 0.530]	0.458 [0.349, 0.566]	D 0.301	all three
Algebra II	cluster tag: recognition or interpretation	57	0.474 [0.351, 0.614]	0.491 [0.368, 0.614]	0.368 [0.246, 0.491]	B 0.263	Qwen only

transcription, and the geometry rule stops firing on 31 items whose transcribed options carry fewer than three numbers). On the text layer, 19 of 27 geometry administrations have a rate above chance (sign test $p = 0.014$), only one clears the item bar on its own, and the rate is the same on faithful (0.323, $n = 99$) and not-faithful (0.326, $n = 178$) items, so the dirt moved n and not the rate; all three EQAO years are above chance by rate and one clears individually, and the $k = 3$ clustered interval is descriptive. The geometry cue is central-ness: upper-central 0.330 ($n = 279$, [0.272, 0.387]) and nearest-to-mean 0.312 also clear, smallest and largest (0.219 each) do not. On EQAO grade 6 only the pre-registered index clears; upper-central is 0.250 [0.125, 0.400], so that witness is specific to the second-smallest option. Both rows are per-cell existence rows on the page-read options that no Holm family keeps; they are replication candidates, and on the 80 and 18 matched items, matching the key does not require the tagged operation.

The other three uncorrected clears are not claimed: Algebra II under the same rule is 0.298 at $n = 372$, with item interval [0.253, 0.344] and clustered interval [0.255, 0.341] sitting on chance; TEKS Algebra I stem-number repeat is 0.395, $n = 43$, [0.256, 0.535]; TEKS grade 5 unit-rate is 0.417, $n = 12$, [0.167, 0.750], chance 0.146. Searched programs that clear the machine bar on TEKS grade 6 and on Regents Algebra I and geometry have n between 11 and 20, 1,847 programs were searched, and all four fail a late-administration held-out test (late $n = 6$ to 10, intervals include chance). Fitted stem n -gram and overlap programs fail every temporal slice (state-pooled 0.257, $n = 1243$) and do not beat chance across corpora (state to TIMSS applying and reasoning 36/122 = 0.295, [0.213, 0.377], chance 0.254); one STAAR refit reaches 0.643 on $n = 14$ from a train n of 8 on one form and is not claimed. Unbound unit-rate, $y = kx$, and percent-of do not clear on TIMSS, PISA, or NAPLAN numeric items. Four catalog cues added after primary scoring (longest option, convergence, stem-word overlap, and mean of other options) give 22 further tests, and the 127-test combined family has no Holm

survivor; the one further per-cell clear, mean of other options on geometry at 0.327 ($n = 156$, [0.256, 0.404]), is not independent of the central-value cue. Over the exploratory 17-cell slice of lower-central alone (Table 8), defined after seeing which rule cleared, Holm keeps EQAO grade 6 and drops geometry.

An item the rule answers: EQAO 2023 grade 6, released questions, printed card page 8 (key table page 21), file tag B1-KU (B1 is EQAO's content code; KU is the knowledge-and-understanding category), id eqao-2023-g6-q1. Stem: "Which number has 4 as its tenths digit?" Options A. 92.174, B. 85.043, C. 79.451, D. 43.256; published key C. Sorted, the numbers are 43.256, 79.451, 85.043, 92.174, and index $\lfloor (4-1)/2 \rfloor = 1$ is 79.451, option C. The program never reads the stem and never inspects a tenths digit. The item is print-faithful, and its page-read options are the four numbers above.

Strategy programs. Three of the 67 strategy tests clear the per-cell bar, and one survives Holm: arithmetic closure on STAAR grade 5, which fires on 40 of 60 selected items and matches 21 keys (0.525, $n = 40$, item 95% CI [0.375, 0.675], chance 0.250, one-sided binomial $p = 0.000175$ against the Holm threshold $0.05/67 = 0.000746$). It is the only Holm reject in the family and in the 172-test sensitivity pool (threshold $0.05/172 = 0.000291$), and it is a discovery-set reject on the frozen 2019/2021/2022 census. It holds under the pre-registered robustness cuts (depth 1 only $17/30 = 0.567$ [0.400, 0.733]; depth 2 without the geometry extras π and $ab/2$ $20/40 = 0.500$ [0.350, 0.650]), and clustered by STAAR year it is [0.455, 0.615] (2019 8/13, 2021 8/16, 2022 5/11; sign test $p = 0.125$ on three administrations, which are the whole cell). Random tie-breaks among minimum-depth options give an expected rate of 0.367 and a 1,000-draw interval of [0.250, 0.475]; the observed 0.525 sits above that interval. Lower-central alone on the same 40 items is $13/40 = 0.325$ [0.200, 0.475] and does not clear; 27 of the 40 picks were settled by the lower-central tie-break among minimum-depth options (15 correct) and 13 by depth alone, and mean closure density is 3.3 of 4 options. The catalog unit-rate row on the

Table 4. Per-cell catalog existence rows for the lower-central rule (pick the second-smallest numeric option) on three layers: the PDF text layer, the print-faithful filter (items whose text layer matches the page), and options re-read from the page image (Section 4.1). n is items the rule fires on, not the cell size in Table 6. Item interval resamples items; clustered interval resamples administrations (year-session for Regents, year for EQAO). Bar: $n \geq 10$ and item lower bound above chance 0.250. Both rows clear on the page-read options and neither survives Holm over 105 a priori tests; they are replication candidates, not discoveries. The EQAO rule is specific to the second-smallest option.

Authority	Claim	Layer	n	Pass	Item 95% CI	Clustered 95% CI	Bar
NY Regents	geometry	text layer	279	0.323	[0.265, 0.380]	[0.265, 0.374]	yes
NY Regents	geometry	print-faithful filter	99	0.323	[0.232, 0.414]	[0.227, 0.414]	no
NY Regents	geometry	page-read options	246	0.325	[0.264, 0.382]	[0.264, 0.387]	yes
EQAO	grade 6	text layer	40	0.500	[0.350, 0.650]	[0.417, 0.588]	yes
EQAO	grade 6	print-faithful filter	14	0.571	[0.286, 0.857]	[0.400, 0.714]	yes, thin
EQAO	grade 6	page-read options	38	0.474	[0.316, 0.632]	[0.273, 0.588]	yes

Table 5. Fixed-order strategy portfolio on the five headline cells. Rate is against effective chance of the branch used. Witnesses in this channel must also exceed the 95th percentile of 200 random programs of matched complexity. No cell clears that bar.

Authority	Claim	n	Pass	Eff. chance	95% CI	p95	Bar
NY Regents	geometry	408	0.279	0.253	[0.238, 0.321]	0.297	no
EQAO	grade 6	53	0.453	0.267	[0.321, 0.585]	0.472	no
TIMSS	knowing	65	0.308	0.271	[0.200, 0.431]	0.292	no
TIMSS	applying	54	0.333	0.254	[0.204, 0.463]	0.370	no
TIMSS	reasoning	68	0.309	0.263	[0.206, 0.426]	0.338	no

Table 6. Minimum detectable bypass rate at this cell’s n (every tagged item, not the subset a given rule scores). Independent is an item-level bootstrap on that n . Clustered is 80% power under administration resampling. TIMSS clustered entries are that resampling on selected-response items ($n = 65/54/68$ for knowing, applying, reasoning), not a rescaling of the independent floors. TIMSS rows pool 2011 and earlier public-release cycles.

Authority	Claim	n	Independent	Clustered
NY Regents	Algebra I	598	0.284	0.303
NY Regents	Algebra II	442	0.290	0.308
NY Regents	geometry	408	0.294	0.309
TIMSS	applying	68	0.353	0.426
TIMSS	knowing	81	0.358	0.415
TIMSS	reasoning	86	0.372	0.441
EQAO	grade 6	53	0.396	0.415

same cell (0.417, $n = 12$) is a different program with smaller n , not a second mechanism. Thirteen of the 40 fired stems contain artifact numerals (axis ticks, thousands separators, stem-and-leaf digits, footer grade numbers, leaked neighboring text), a construct problem for those 13; the remaining 27, a population defined after the fact, are $16/27 = 0.593$ [0.370, 0.778], and unique depth-1 closure is the key on 6 of the 10 items where exactly one option is in the depth-1 set.

Geometry closure is 0.196 ($n = 224$, [0.143, 0.250]) and does not clear.

The registered replication prediction was that filtered closure on held-out grade 5 items alone would have a lower 95% CI strictly above 0.250. On the version-1 unrepaired parse, 18 of 44 held-out keys match (0.409, [0.273, 0.545]): the prediction is met at $n = 44$, below the registered power floor of 46. After a mechanical last-option page-leak repair hashed before rescoring (42 last-option repairs, 3 stacked-fraction rejoins; hand check 14 agree, 4 disagree, 2 uncertain, 0 key mismatches), 17 of 44 match (0.386, [0.250, 0.545]), an underpowered miss. English grade 5 forms for 2017 and 2018 were then parsed (the 2015 key and the 2020 and 2023 tests were not found): new forms alone are $4/20$ (0.200, [0.050, 0.400]), and combined repaired plus new is $21/64 = 0.328$, [0.219, 0.438], at $n = 64 \geq 46$. The Holm-surviving row on the frozen discovery cell did not replicate on held-out 2013/2014/2016/2017/2018 administrations, and that null is powered. Grade 4 lower-central is $10/17$ after repair, 5 of the hits are parse artifacts, and the clean $5/9$ does not clear.

The distractor-derivation hub never clears against the 95th percentile of random programs (geometry 0.255, $n = 161$, [0.193, 0.317] versus p95 0.311; EQAO grade 6 0.321, $n = 28$, versus 0.438; TEKS grade 4 0.455, $n = 22$, [0.273, 0.682] versus 0.476); the unique-hub share among 992 family-scored items is 0.189, and most hubs are ties broken by lower-central. Constraint elimination barely eliminates: mean remaining options stay near 4 (geometry 3.92, EQAO grade 6 3.77; share

Table 7. Method control: the same 1,847-program searcher recovers published options-only or question-ignored rates on OpenBookQA and SWAG. Chance is 0.25 except CommonsenseQA (0.20, five options). RACE and CommonsenseQA do not yield a witness here.

Corpus	Published cue number	Recovered	n	Witness
OpenBookQA	0.496, question ignored [Mihaylov et al. 2018]	0.450 (longest option)	329	yes
SWAG	0.436, ending-only [Zellers et al. 2018]	0.565 (searched)	62	yes
RACE [Lai et al. 2017]	(no official options-only number)	0.375	8	no ($n < 10$)
CommonsenseQA [Talmor et al. 2019]	chance 0.20	0.227	22	no

of options removed 0.025 and 0.019), key survival is 0.985 on geometry, 0.906 on EQAO grade 6, and 0.877 on TEKS grade 8, two specified intersections kill the published key (probability with percent-point options; interior angle in $(0, 180)$ on a rectangle angle sum), a hand check of 15 keyword-matched items keeps 13 keys, and its two per-cell clears, geometry 0.307 ($n = 404$, $[0.262, 0.351]$ versus effective chance 0.254) and EQAO grade 6 0.412 ($n = 51$, $[0.275, 0.529]$ versus 0.268), are the lower-central tie-break again (the keyword-matched geometry subset still clears at 0.352, $n = 162$; the EQAO keyword subset, $n = 19$, does not) and add nothing to the catalog rows. The portfolio never clears the random-program 95th percentile (Table 5); EQAO grade 6 clears chance and one-over- k (0.453, $n = 53$, $[0.321, 0.585]$ versus 0.253) and fails p95 0.472, with almost all mass on lower-central among survivors. On the OpenBookQA and SWAG hold-out splits of Table 7, elimination and portfolio sit on chance (OpenBookQA elimination 0.260 $[0.218, 0.302]$, $n = 400$, portfolio 0.268 $[0.225, 0.310]$; SWAG elimination 0.248 $[0.205, 0.288]$, portfolio 0.240 $[0.198, 0.280]$), and the numeric hub, closure, and backsolve do not fire on language items.

Image-channel backsolving. Text-layer backsolving produces 6 unique fires among 1,186 solving-tagged stems and no cell with $n \geq 10$; most equations in the source PDFs are images. The image addendum fired on 17 of 598 Algebra I solving-tagged selected-response items; two fires are recoded as abstentions (an invented formula with no displayed equation; an incomplete two-equation extract), and of the remaining 15, 14 match the key (0.933, $n = 15$, item 95% CI $[0.800, 1.000]$, chance 0.25). The miss is a correctly transcribed equation on an item that asks which point is not on the graph. Coverage is $15/598 = 0.025$, no other cell reaches 10 fires, and the row reads the printed equation from the page image, so it does not depend on the text layer. On those 14 items a pass does not require solving; the row is an existence measurement outside the Holm families, not a census of Algebra I.

4.4 TIMSS and the controls

No catalog, strategy, or partial-input program clears its bar on TIMSS knowing, applying, or reasoning. Lower-central on TIMSS reasoning is 0.368, $n = 57$, $[0.246, 0.491]$ (Table 8);

elimination on reasoning is 0.379 ($n = 66$, $[0.258, 0.500]$ versus effective chance 0.263); the portfolio rows are in Table 5; the masked-stem rows are in Table 9, where Qwen 14B knowing (0.591, $n = 22$) rejects Holm against chance and does not beat the population modal 0.409, and the first-run knowing rate 0.657 ($n = 35$) was carried by items where nothing was withheld. TIMSS 2011 is mostly removed from the language-model population by country-table contamination (17 items kept).

That null is power-limited. Table 6 records, for every cell with $n \geq 10$, the smallest observed pass rate whose item-level bootstrap interval excludes chance at the cell’s tagged-item count (the independent-items floor) and the bypass rate with 80% power to clear the bar when administrations are resampled (the clustered floor). Twenty-one cells have $n \geq 10$; the smallest independent floor is 0.284 on Algebra I ($n = 598$), whose clustered floor is 0.303. The TIMSS cells pool the 2011 cycle with earlier released cycles at $n = 81/68/86$ tagged and 65/54/68 selected-response items; their independent floors are 0.358/0.353/0.372 and their clustered floors 0.415/0.426/0.441. A bypass rate in the low thirties on those cells would not have been seen.

The null is not a failure of the searcher. On 1,200 synthetic four-option items, each planted atomic program scored 50/50 on holdout with confidence interval $[1, 1]$, recall of planted flaw types is 6/6, and the false-positive rate on matched clean holdout is 0; that shows the searcher fires when a unique surface cue is planted, not that released items contain one. On public cue sets (Table 7), the same 1,847-program space, fit on a train split and scored held-out, recovers documented shallow cues on OpenBookQA and SWAG, two corpora whose authors published an options-only or question-ignored number.

5 Limitations

The instrument that reads the page is a vision-language model, and 175 of the 606 algebra items did not verify under two readings by Qwen2-VL-7B-Instruct. A third reading by a different family, LLaVA-NeXT Vicuna 13B, agrees with the first Qwen reading on 160 items, 153 of them in the claimed population, so on that subset a reading error would have to be shared by two model families rather than made twice by one; on it the Algebra I rows hold on all three scorers and the

Algebra II rows on the two Qwen scorers, and the Algebra II Phi-4 row does not clear at $n = 57$. What remains: 446 items do not verify across families, 5 of them two-page crops that exceeded the LLaVA context, so the claimed population is still mostly items two readings of one model agree on; only 20 items were checked against the page by a human; a hard-to-read item is more likely to be outside every verified set; and the two catalog rows rest on a single reading of their options. A human transcription of the 175 items that fail the Qwen check would let every row be reported on the full 606.

What an Algebra I item asks is measured by two language-model coders and by the publisher’s tag, and the three disagree: the coders agree with each other at $\kappa = 0.576$ and with the cluster split at 0.071 and 0.230. No human coded the items, so a disagreement between coder and tag can be a coder error as easily as a tag error; automated tagging of item demand has that record elsewhere, where language-model Q-matrices for a reading test agreed with a validated expert Q-matrix at κ up to 0.63, with wide variation across models [Fan and Bialo 2026]. This limits the stronger statement: whether a pass on items that require operating on the givens omits that operation rests on the both-coders execution subset ($n = 142$ verified), on which Phi-4 does not clear, and the subset itself moves with the coder. A human coding of the 358 Algebra I items by two trained raters under the same rubric, with agreement reported, would fix the classes and is the measurement that would settle the question.

The template-memorization explanation is untested. The option-numeral control rules out recall of specific values, and the flat administration-year probe speaks against a template that drifted, but a reader that had learned the recurring stem-and-option templates of Regents items would keep its rate under a change of values exactly as the form component does. That leaves open whether the form component is a property of the item form or of the training corpus. The test is novel items written to the same templates that no model has seen, in the manner of a benchmark rewritten in the style of a public one [Zhang et al. 2024], scored under the same masking.

The Qwen 14B text-layer isomorph arms are below the reproduction floor. Re-scored original letters agreed with the saved letters on $592/606 = 0.977$ against a 0.98 floor, twice, and the cause is a logit shift between the torch version of the saved run and the torch version of both re-runs, which agree with each other to the last bit: six of the 14 changed items are exact ties on the re-run, an fp32 logits head recovers four, and one item moves by several nats (Appendix A.6). The arms remain excluded and are shown as audit trail; the repaired-layer control covers Qwen 14B from a single model load and is the control we claim, so the gap affects only that trail. A pre-registered margin-based floor, or a re-run of the text-layer arms from one load, would close it.

The human masked-stem sheet has one rater on 40 items, and its interval $[0.225, 0.525]$ is uninformative. Whether a

human denied the given quantities reads the same form cue as the models is unknown, and the language-model finding is stated as a fact about models for that reason. Several raters on a sheet sized to detect the model rate against chance would answer it.

The EQAO grade 6 and geometry lower-central rows are small and unreplicated. EQAO grade 6 fires on 38 page-read items from three administrations; geometry fires on 246 items, and on the text layer 19 of 27 administrations are above chance but only one clears on its own; neither survives Holm over the 105 a priori tests. Both are existence rows, and a reader should hold them as hypotheses about the second-smallest option rather than as census findings. A pre-registered replication on administrations outside the census, EQAO releases after 2023 and Regents geometry forms not yet ingested, is the test.

The STAAR closure replication is limited to the forms we could fetch and to a parse we checked by hand on 20 items. The Holm-surviving discovery row replicated as an underpowered miss on 44 held-out grade 5 items and failed at $n = 64$ once the 2017 and 2018 English forms were added; the 2015 key and the 2020 and 2023 tests were not found, the parse repair’s hand check disagreed on 4 of 20, and 13 of the 40 discovery stems carry artifact numerals. The powered null therefore rests on those forms and that parse. Fetching the missing forms and hand-verifying the parse on all 64 replication items would make the null a clean one.

The instruction-only family, a retriever of worked examples from the same unit with no rule induction, is specified in the protocol and was not run, because curriculum scoring keys were not in the public files we could ingest. That restriction kind, the template imitator, is therefore untested, and the census tests three of the four specified families. An open curriculum whose unit assessments and keys are public would let it run.

Constructed-response items are scored only where a scoring guide reduces to a value. The 160 numeric-answer items were scored by the unbound-formula rules alone, explanations were never graded, and the constructed-response parts of each Regents administration were not ingested, so every claimed row is a fact about the selected-response part of the exam. An executor of the published rating guide, or value matching on the numeric-answer parts as a first channel, would extend the search to the rest of the form.

The ingestion is incomplete. Most NYSED 2018 graphic stems and the TIMSS 2007 release are out of this file, and earlier English STAAR forms entered only as a replication set. Those gaps lower the cell sizes, and with them the TIMSS power floors of Table 6; ingesting them is mechanical.

6 Conclusion

We asked whether a pass on a tagged assessment item requires the tagged competency, and tested it without students by searching 2,405 public-release mathematics items for assessment bypasses under a pre-registered bar. Language models denied every given quantity recover the New York Regents Algebra I and Algebra II keys in two model families, and the recovery survives a form-preserving change of every option value. Catalog rules denied the tagged operation match keys on two cells at sizes that make them replication candidates. The one strategy program that survived Holm on its discovery cell did not replicate on held-out administrations. The TIMSS cognitive-domain cells yield no bypass at their public sizes. Every claimed rate is measured on items re-read from the printed page, which the released text layer often misstates.

On the items a bypass matches, a correct answer is read as evidence of an operation on quantities the solver never saw, so the key does not carry the evidence it is read as. That is a fact about the item and its scoring rule, and it can be checked before any student sits the test.

The search needs the tag text, the item text, the keys, an executable scoring rule, and a machine-enforceable restriction on the solver, by instruction set or by withholding part of the item; where those are public, it runs as it did here. Where they are not, the same search waits on a published scoring rule and a restriction a machine can enforce, and the census here is a lower bound on what that search finds. The one-sided scope of the claim is stated once with the bypass definition (Section 3); it is not repeated here.

References

- Aishwarya Agrawal, Dhruv Batra, and Devi Parikh. 2016. Analyzing the Behavior of Visual Question Answering Models. In *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing*, Jian Su, Kevin Duh, and Xavier Carreras (Eds.). Association for Computational Linguistics, Austin, Texas, 1955–1960. doi:10.18653/v1/D16-1203
- American Educational Research Association, American Psychological Association, and National Council on Measurement in Education. 2014. *Standards for Educational and Psychological Testing*. American Educational Research Association, Washington, DC.
- Yigal Attali and Maya Bar-Hillel. 2003. Guess Where: The Position of Correct Answers in Multiple-Choice Test Items as a Psychometric Variable. *Journal of Educational Measurement* 40, 2 (2003), 109–128. doi:10.1111/j.1745-3984.2003.tb01099.x
- Nishant Balepur, Abhilasha Ravichander, and Rachel Rudinger. 2024. Artifacts or Abduction: How Do LLMs Answer Multiple-Choice Questions Without the Question?. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Lun-Wei Ku, Andre Martins, and Vivek Srikumar (Eds.). Association for Computational Linguistics, Bangkok, Thailand, 10308–10330. doi:10.18653/v1/2024.acl-long.555
- Nishant Balepur and Rachel Rudinger. 2024. Is Your Large Language Model Knowledgeable or a Choices-Only Cheater?. In *Proceedings of the 1st Workshop on Towards Knowledgeable Language Models (KnowLLM 2024)*, Sha Li, Manling Li, Michael JQ Zhang, Eunsol Choi, Mor Geva, Peter Hase, and Heng Ji (Eds.). Association for Computational Linguistics,

- Bangkok, Thailand, 15–26. doi:10.18653/v1/2024.knowllm-1.2
- Nishant Balepur, Rachel Rudinger, and Jordan Boyd-Graber. 2025. Which of These Best Describes Multiple Choice Evaluation with LLMs? A) Forced B) Flawed C) Fixable D) All of the Above. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Wanxiang Che, Joyce Nabende, Ekaterina Shutova, and Mohammad Taher Pilehvar (Eds.). Association for Computational Linguistics, Vienna, Austria, 3394–3418. doi:10.18653/v1/2025.acl-long.169
- Jimmy de la Torre and Chia-Yi Chiu. 2016. A General Method of Empirical Q-matrix Validation. *Psychometrika* 81, 2 (2016), 253–273. doi:10.1007/s11336-015-9467-8
- Kevin Fan and Jacquelyn A. Bialo. 2026. The Use of AI Tools to Develop and Validate Q-Matrices. In *Proceedings of the 90th Annual International Meeting of the Psychometric Society*. Psychometric Society. doi:10.64028/imps2025.whsmqf4ego
- Qixiang Fang, Daniel Oberski, and Dong Nguyen. 2025. PATCH! Psychometrics-AssisTed BenCHmarking of Large Language Models against Human Populations: A Case Study of Proficiency in 8th Grade Mathematics. In *Proceedings of the Fourth Workshop on Generation, Evaluation and Metrics (GEM²)*. Association for Computational Linguistics, Vienna, Austria and virtual meeting, 808–823. https://aclanthology.org/2025.gem-1.68/
- Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A. Wichmann. 2020. Short-cut learning in deep neural networks. *Nature Machine Intelligence* 2, 11 (Nov. 2020), 665–673. doi:10.1038/s42256-020-00257-z
- Suchin Gururangan, Swabha Swayamdipta, Omer Levy, Roy Schwartz, Samuel R. Bowman, and Noah A. Smith. 2018. Annotation Artifacts in Natural Language Inference Data. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 2 (Short Papers)*, Marilyn Walker, Heng Ji, and Amanda Stent (Eds.). Association for Computational Linguistics, New Orleans, Louisiana, 107–112. doi:10.18653/v1/N18-2017
- Thomas M. Haladyna, Steven M. Downing, and Michael C. Rodriguez. 2002. A Review of Multiple-Choice Item-Writing Guidelines for Classroom Assessment. *Applied Measurement in Education* 15, 3 (July 2002), 309–333. doi:10.1207/s15324818ame1503_5
- Thomas M. Haladyna and Michael C. Rodriguez. 2013. *Developing and Validating Test Items*. Routledge, New York. doi:10.4324/9780203850381
- Paul W. Holland and Dorothy T. Thayer. 1986. Differential Item Functioning and the Mantel-Haenszel Procedure. *ETS Research Report Series* 1986, 2 (1986). doi:10.1002/j.2330-8516.1986.tb00186.x
- International Association for the Evaluation of Educational Achievement. 2009. *TIMSS 2011 Assessment Frameworks*. Technical Report. TIMSS & PIRLS International Study Center, Boston College, Chestnut Hill, MA. https://timssandpirls.bc.edu/timss2011/frameworks.html
- Michael T. Kane. 1992. An argument-based approach to validity. *Psychological Bulletin* 112, 3 (Nov. 1992), 527–535. doi:10.1037/0033-2909.112.3.527
- Michael T. Kane. 2013. Validating the Interpretations and Uses of Test Scores. *Journal of Educational Measurement* 50, 1 (2013), 1–73. doi:10.1111/jedm.12000
- Irvin R. Katz, Randy Elliot Bennett, and Aliza E. Berger. 2000. Effects of Response Format on Difficulty of SAT-Mathematics Items: It’s Not the Strategy. *Journal of Educational Measurement* 37, 1 (2000), 39–57. doi:10.1111/j.1745-3984.2000.tb01075.x
- Divyansh Kaushik and Zachary C. Lipton. 2018. How Much Reading Does Reading Comprehension Require? A Critical Investigation of Popular Benchmarks. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, Ellen Riloff, David Chiang, Julia Hockenmaier, and Jun’ichi Tsujii (Eds.). Association for Computational Linguistics, Brussels, Belgium, 5010–5015. doi:10.18653/v1/D18-1546
- Guokun Lai, Qizhe Xie, Hanxiao Liu, Yiming Yang, and Eduard Hovy. 2017. RACE: Large-scale ReAding Comprehension Dataset From Examinations. In *Proceedings of the 2017 Conference on Empirical Methods in Natural*

- Language Processing, Martha Palmer, Rebecca Hwa, and Sebastian Riedel (Eds.). Association for Computational Linguistics, Copenhagen, Denmark, 785–794. doi:10.18653/v1/D17-1082
- Li Lucy, Tal August, Rose E. Wang, Luca Soldaini, Courtney Allison, and Kyle Lo. 2024. Mathfish: Evaluating Language Model Math Reasoning via Grounding in Educational Curricula. arXiv:2408.04226 [cs.CL] <https://arxiv.org/abs/2408.04226>
- Inbal Magar and Roy Schwartz. 2022. Data Contamination: From Memorization to Exploitation. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, Smaranda Muresan, Preslav Nakov, and Aline Villavicencio (Eds.). Association for Computational Linguistics, Dublin, Ireland, 157–165. doi:10.18653/v1/2022.acl-short.18
- R. Thomas McCoy, Ellie Pavlick, and Tal Linzen. 2019. Right for the Wrong Reasons: Diagnosing Syntactic Heuristics in Natural Language Inference. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, Anna Korhonen, David Traum, and Lluís Màrquez (Eds.). Association for Computational Linguistics, Florence, Italy, 3428–3448. doi:10.18653/v1/P19-1334
- Samuel Messick. 1995. Validity of psychological assessment: Validation of inferences from persons’ responses and performances as scientific inquiry into score meaning. *American Psychologist* 50, 9 (Sept. 1995), 741–749. doi:10.1037/0003-066x.50.9.741
- Todor Mihaylov, Peter Clark, Tushar Khot, and Ashish Sabharwal. 2018. Can a Suit of Armor Conduct Electricity? A New Dataset for Open Book Question Answering. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, Ellen Riloff, David Chiang, Julia Hockenmaier, and Jun’ichi Tsujii (Eds.). Association for Computational Linguistics, Brussels, Belgium, 2381–2391. doi:10.18653/v1/D18-1260
- Jason Millman, Carol H. Bishop, and Robert Ebel. 1965. An Analysis of Test-Wiseness. *Educational and Psychological Measurement* 25, 3 (Oct. 1965), 707–726. doi:10.1177/001316446502500304
- Seyed Iman Mirzadeh, Keivan Alizadeh, Hooman Shahrokhi, Oncel Tuzel, Samy Bengio, and Mehrdad Farajtabar. 2025. GSM-Symbolic: Understanding the Limitations of Mathematical Reasoning in Large Language Models. In *The Thirteenth International Conference on Learning Representations (ICLR)*. <https://openreview.net/forum?id=AjXkRZlvjB>
- Steven Moore, Eamon Costello, Huy A. Nguyen, and John Stamper. 2024. An Automatic Question Usability Evaluation Toolkit. In *Artificial Intelligence in Education (AIED 2024), Lecture Notes in Computer Science*. Springer Nature Switzerland, 31–46. doi:10.1007/978-3-031-64299-9_3
- Adam Poliak, Jason Naradowsky, Aparajita Haldar, Rachel Rudinger, and Benjamin Van Durme. 2018. Hypothesis Only Baselines in Natural Language Inference. In *Proceedings of the Seventh Joint Conference on Lexical and Computational Semantics*, Malvina Nissim, Jonathan Berant, and Alessandro Lenci (Eds.). Association for Computational Linguistics, New Orleans, Louisiana, 180–191. doi:10.18653/v1/S18-2023
- Oscar Sainz, Jon Campos, Iker García-Ferrero, Julen Etxaniz, Oier Lopez de Lacalle, and Eneko Agirre. 2023. NLP Evaluation in trouble: On the Need to Measure LLM Data Contamination for each Benchmark. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, Houda Bouamor, Juan Pino, and Kalika Bali (Eds.). Association for Computational Linguistics, Singapore, 10776–10787. doi:10.18653/v1/2023.findings-emnlp.722
- Alon Talmor, Jonathan Herzig, Nicholas Lourie, and Jonathan Berant. 2019. CommonsenseQA: A Question Answering Challenge Targeting Commonsense Knowledge. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, Jill Burstein, Christy Doran, and Tamar Solorio (Eds.). Association for Computational Linguistics, Minneapolis, Minnesota, 4149–4158. doi:10.18653/v1/N19-1421
- Kikumi K. Tatsuoka. 1983. Rule Space: An Approach for Dealing with Misconceptions Based on Item Response Theory. *Journal of Educational Measurement* 20, 4 (1983), 345–354. doi:10.1111/j.1745-3984.1983.tb00212.x
- Ragip Terzi and Sedat Sen. 2019. A Nondiagnostic Assessment for Diagnostic Purposes: Q-Matrix Validation and Item-Based Model Fit Evaluation for the TIMSS 2011 Assessment. *Sage Open* 9, 1 (2019), 2158244019832684. doi:10.1177/2158244019832684
- Patrick Watson, TengFei Ma, Ravi Tejjwani, Maria Chang, JaeWook Ahn, and Sharad Sundararajan. 2018. Human-level Multiple Choice Question Guessing Without Domain Knowledge: Machine-Learning of Framing Effects. In *Companion Proceedings of the The Web Conference 2018 (WWW ’18 Companion)*. ACM, 299–303. doi:10.1145/3184558.3186340
- A. P. White and J. E. Zammarelli. 1981. Convergence Principles: Information in the Answer Sets of Some Multiple-Choice Intelligence Tests. *Applied Psychological Measurement* 5, 1 (1981), 21–27. doi:10.1177/014662168100500103
- Rowan Zellers, Yonatan Bisk, Roy Schwartz, and Yejin Choi. 2018. SWAG: A Large-Scale Adversarial Dataset for Grounded Commonsense Inference. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, Ellen Riloff, David Chiang, Julia Hockenmaier, and Jun’ichi Tsujii (Eds.). Association for Computational Linguistics, Brussels, Belgium, 93–104. doi:10.18653/v1/D18-1009
- Hugh Zhang, Jeff Da, Dean Lee, Vaughn Robinson, Catherine Wu, Will Song, Tiffany Zhao, Pranav Raja, Charlotte Zhuang, Dylan Slack, Qin Lyu, Sean Hendryx, Russell Kaplan, Michele Lunati, and Summer Yue. 2024. A Careful Examination of Large Language Model Performance on Grade School Arithmetic. In *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*. 46819–46836. doi:10.52202/079017-1485

A Implementation details

This appendix records the settings that fix the reported numbers. Nothing here changes a claim in Section 3.

A.1 Frozen strategy programs

The hash of the frozen rule file, recorded 2026-09-16 20:36:16 UTC before any pass rate was computed, is a0eda4bdb02e54765bb7b49dca5e2e8ea9cd9ebb66de67b2aae074e7d555f0b4.

The distractor-derivation hub builds a directed graph on option values under a fixed error-transform set and picks the unique maximum out-degree, else lower-central among ties. Constraint elimination uses shallow stem keywords: nonnegative counts, probability in $[0, 1]$, percent in $[0, 100]$, positive magnitudes, angle ranges, and mean or median inside the listed data. Arithmetic closure at depth two is unbound execution of $+$, $-$, $\times$, $\div$, squares, and square roots on up to 12 stem numbers, with π and $ab/2$ on geometry-word stems; it picks the option that first appears, else lower-central among that depth. Backsolving parses a one- or two-variable equation from the stem and picks the unique option that satisfies it. The portfolio runs those four in a fixed order with lower-central among survivors and letter B last.

The machine bar for hub and portfolio is the 95th percentile of 200 random programs of matched complexity, seed 20260916.

The image-channel backsolving addendum was hashed before scoring; the sha256 of its preregistration is d6ee8cafcd35d04bba7b7fd4d4d364ee6f274d715e9b6b7020c454dbdfbd994732. It uses Qwen2-VL-7B-Instruct (Hugging Face

revision eed13092ef92e448dd6875b2a00151bd3f7db0ac) to transcribe the displayed equation from a page image; substitution is symbolic, outside the model.

A.2 Language-model scoring

The scored letter is the option letter with the highest next-token log-probability after the chat template, taking the maximum of the bare letter token and the space-prefixed letter token. The seed is 20260916.

Models and Hugging Face revisions:

- Qwen2.5-7B-Instruct: a09a35458c702b33eeacc393d103063234e8bc28
- Qwen2.5-14B-Instruct: cf98f3b3bbb457ad9e2bb7ba f9a0125b6b88caa8
- Mistral-7B-Instruct-v0.3: c170c708c41dac9275d15a8fff4eca08d52bab71
- Phi-4 (microsoft/phi-4): 2db69c1c3e91a05d2c64a3185acfbaf36f744e25

The clean-option filter is mechanical. It keeps 1,496 of 2,245 selected-response items. Of those, 1,487 have a single option-letter key and are scored. The masker withheld nothing on 272 of those 1,487, leaving the primary population of 1,215 items with at least one withheld quantity. The first-run pooled rates on all 1,487 are a reference only.

The masking rules were hashed before scoring. Version 1 is af473bb6cd9d46dc74e034f4460d56a808273fe83ffa457cd7d122c40ae80e8. Version 2 is 8e6186728bf247d8349f1d94b68280b49eeebac7684c7ebeaf80f29345a43ae5.

Option-numeral isomorph control. The preregistration of this control was hashed before any scoring for it: 997242b9fb7437ffc2e86e2759e1373ba8ad98e231b0836ef8798a2266523e9b. Population: the 606 withheld-quantity Algebra I and Algebra II items under masking version 1 (358 and 248), with the subsets of Appendices A.4 and A.5. Seed 20260916; per item and arm the generator is seeded from the sha256 of seed, arm, and item id, so the substitution does not depend on processing order. Numerals in the option strings are integers (including coefficients and exponents), decimals, comma-grouped integers, and fractions. A replacement keeps the digit count, sign, number of decimal places, leading-dot form, and comma grouping, keeps denominators nonzero, and maps every repeat of the same literal within an item to the same new literal; zero becomes a positive one-digit integer. Variables, operators, units, words, option letters, letter order, and the key letter are unchanged. In the rank-preserving arm the leading numbers of the options are redrawn until their pairwise order equals the original order, ties remaining ties; the scrambled arm has no order constraint. Items with no numeral in any option (61 Algebra I, 21 Algebra II) are unchanged under both arms, are included in pass rates, and are excluded from the paired difference. Example, June 2016 Algebra I item 7: options A to D 6.2, 7.3, 8.6, 8.8 become 7.4, 9.6, 9.7, 9.8 in the rank-preserving arm and 8.2, 4.9, 4.2,

2.7 in the scrambled arm. These invariants were asserted on every item before any perturbed arm was scored. Scoring reuses the masked-stem letter rule above, one model load per model. Original letters are re-scored, not copied, and must agree with the saved per-item predictions on at least 0.98 of the 606 ids before the perturbed arms are scored: Qwen2.5-7B-Instruct 595/606 = 0.982 and Phi-4 606/606 = 1.000 cleared; Qwen2.5-14B-Instruct 592/606 = 0.977 did not, so its perturbed arms were not scored. Pass-rate intervals are percentile bootstraps seeded by $21 + n$ and paired differences by $31 + n$, 1,000 replicates each. The pre-registered reading applies to Algebra I print-faithful items on the rank-preserving arm, each model separately: form-based if the perturbed rate clears the channel bar and the paired difference includes zero; not a form effect if the perturbed rate falls to chance; mixed otherwise. Under that reading, on the text layer, Phi-4 is form-based and Qwen 7B is mixed (Table 14). The same perturbation, rank-preserving arm only, was then applied to the repaired options of Appendix A.6 under the preregistration of that layer; there the original and the perturbed letters were scored in the same run, so no reproduction floor applies. On that layer Qwen 7B's paired difference includes zero on both cells and Qwen 14B's and Phi-4's exclude it on Algebra I (Table 2). Four repaired items whose perturbation raised or exceeded a 20-second limit are excluded from the perturbed arm and counted.

A.3 Searched program class

The feature library holds eleven surface cues, enumerated as conjunctions and signed weights up to complexity 3, plus greedy and evolutionary search: 1,847 programs logged.

A.4 Print-fidelity check

Population: the 955 items in the five cells named in Section 3.6 (Algebra I masked-stem 358, Algebra II masked-stem 248, geometry lower-central fired 279, EQAO grade 6 lower-central fired 40, geometry masked-stem 166; 136 geometry items are in both geometry cells). Transcriber: Qwen2-VL-7B-Instruct, revision eed13092ef92e448dd6875b2a00151bd3f7db0ac, bf16, greedy, 640 new tokens, seed 20260916, asked only to transcribe the stem and options of the numbered item from a 150 DPI render of the source PDF page (Regents: a crop from the question heading to the next heading; EQAO: the full page). The preregistration, with the prompt, the page locator, the normalization, and the thresholds, was hashed before any audited or population item was scored: 0211585e9361d3771e04df5e118050ede761971da286f0b1333a7aac289737d6.

Normalization is identical on both sides: Unicode NFKC, LaTeX commands removed except trigonometric and logarithm names, subscripts and braces removed, unit words to symbols, a space at every letter-digit boundary, alphanumeric tokens, and the tokens for square root, π , percent,

degree, and minus dropped. Control characters and operator glyphs therefore do not count; a digit standing in for an operator does. Stem score is multiset token precision, recall, and F1; option score is 1 for equal compact strings, else the maximum of token F1 and a character-level sequence ratio. Faithful: stem F1, precision, and recall each at least 0.8, every option at least 0.9, option counts equal. Unverified: unparseable output, no options, or empty stem. Calibration on the 20 hand-audited items: 17 of 20 agree on faithful versus not, against a pre-registered floor of 16 (the declared threshold grid gave 16 or 17 at every point), so the thresholds were kept. Census verdicts: 434 faithful, 502 not faithful, 19 unverified. Faithful fraction, faithful over faithful plus not faithful: Algebra I 0.523, Algebra II 0.481, geometry lower-central 0.357, EQAO grade 6 0.368, geometry masked-stem 0.657. Two post-hoc checks are disclosed and reported alongside the pre-registered subset: a strict variant that also drops any faithful item whose text layer carries a digit token absent from the transcription (85 of 434), and a locator repair that re-transcribed 12 Regents items and changed three verdicts; neither changes which rows clear, except that EQAO grade 6 does not clear on the strict subset (5/10). Rescoring reruns the catalog rule and reuses the saved per-item language-model predictions with the same bootstraps as the main text (item bootstrap seed 11, 1,000 replicates; cluster bootstrap seed 20260916, 2,000 replicates; language-model percentile bootstrap seeded by $21 + n$, 1,000 replicates). Wall time on one GPU: 26 minutes for the census.

A.5 Standards split

Tags: the “Map to the Learning Standards” table of the NYSED rating guide for each of the 58 Algebra I and Algebra II administrations in the masked-stem primary population, read from the PDF text layer (50 guides, 518 items), from a glyph table for three guides whose Calibri subset has a broken character map (30 items), or by template matching of rendered glyphs against six text-layer guides for five guides with no character map at all (57 items; 0 errors on 33,704 held-out characters). Every parsed table is validated (questions 1 to N contiguous, exactly 24 multiple-choice rows). All 358 Algebra I and 248 Algebra II items carry a cluster code; 33 distinct clusters are used. Cluster headings and standards were fetched from the CCSS site and stored with page hashes before classification. The verb-class rule, vocabulary, and classification of all 33 clusters were hashed before any prediction was joined to a tag: c87a5ae0d897b389dbf5cda52086c1754955f7cbb5f9e7a7382f5ac7cc475488.

Rule: each imperative verb heading a top-level clause of the cluster heading is execution (add, apply a rule, build, calculate, complete the square, compute, construct an equation or model, create, derive, evaluate an expression, factor, find, fit, graph, perform arithmetic operations, produce, rearrange, represent as a graph or table, rewrite, solve, summarize data, transform, translate between forms, write) or recognition

or interpretation (analyze, choose, compare, construct an argument, decide, describe, distinguish, evaluate a process or claim, explain, identify, interpret, justify, know, make an inference, prove, reason, recognize, relate, understand). A purpose clause “use X to V ” takes the class of V ; “extend” and “use X ” with no purpose verb are unclassified, and a heading with no classifiable verb takes the class of its non-plus numbered standards. All execution: execution; all recognition: recognition or interpretation; both: mixed. Algebra I: execution 164 items (clusters A-APR.A, A-CED.A, A-REI.B, A-REI.C, A-REI.D, A-SSE.B, F-BF.A, F-BF.B), recognition or interpretation 151 (A-APR.B, A-REI.A, A-SSE.A, F-IF.A, F-IF.B, F-IF.C, F-LE.B, N-RN.B, S-ID.C), mixed 43 (F-LE.A, N-Q.A, S-ID.A, S-ID.B). Algebra II: 120, 93, and 35. A disclosed stricter execution class, which also requires that no numbered standard under the cluster be purely recognition, has 133 Algebra I items (66 print-faithful) and does not clear on any model on the text layer either. The text-layer and filtered rows use the saved per-item predictions and the same bootstrap as the main text; the full-cell and faithful rows reproduce the text-layer and print-faithful rows exactly. On the repaired layer the same class table is joined to the repaired scores: Algebra I verified 130 execution, 105 recognition or interpretation, 26 mixed; Algebra II verified 83, 57, and 27.

A.6 Repaired layer

The preregistration of the repaired layer and of the Qwen 14B amendment below was hashed before the verification transcription and before any scoring on that layer: 87ec5754c1b696b25366547793365610a13084f1230247b898f564ea9f92caee. Population: the 606 withheld-quantity Algebra I and Algebra II items (358 and 248). Transcription 1 is the stored output of the print-fidelity check (Qwen2-VL-7B-Instruct, revision eed13092ef92e448dd6875b2a00151bd3f7db0ac, prompt and crops of Appendix A.4); items whose stored output has no parseable stem or fewer than two options (15) are left without a repaired stem, and no text is invented. Transcription 2 is a second pass of the same model and revision at greedy decoding, 640 new tokens, seed 20260916, with two deliberate differences: the crop is expanded by 6 pt at the top and bottom (8 pt pad to 14 pt) and the prompt is reworded (it asks the model to copy the printed question, to skip headers, footers, other questions, and the computations box, to copy table cells row by row, to put no answer choice in the stem, and to return one JSON object). Parser, option-label map, and stem post-processing are the functions of the print-fidelity check. An item is repaired-verified when `score_fidelity`, applied with transcription 1 as the item text and transcription 2 as the page reading, returns faithful (stem F1, precision, and recall each at least 0.8; every option at least 0.9; equal option counts). The key letter is the census key. Masking is version 1 (hash above) applied to the transcription-1 stem; options are not masked; the withheld-quantity population is items with a usable transcription 1, the census key among

the transcribed options (no item fails this), and at least one masked token (17 items fail this).

Counts: 431 verified and 175 unverified of 606. Against the filter verdicts, verified items were 246 faithful and 185 not faithful; unverified items were 53 faithful, 107 not faithful, and 15 unverified. Algebra I: 263 verified (150 faithful, 113 not faithful) and 95 unverified; Algebra II: 168 verified (96, 72) and 80 unverified. Withheld-quantity population: 574 (343 Algebra I, 231 Algebra II), of which 261 and 167 verified. Scoring: the models, revisions, seed, prompt mode, and letter rule of Appendix A.2 (Qwen 7B batch size 4; Qwen 14B and Phi-4 batch size 1), on CUDA, one model load per model; the rank-preserving isomorph arm was scored in the same load. Intervals: random. Random. randrange percentile bootstrap, 1,000 replicates, seed $21+n$; paired differences seed $31+n$. Bar: $n \geq 10$, lower bound strictly above 0.25 and strictly above the subset modal-letter frequency. The preregistration’s honesty clause requires the README to say in its first section if a repaired-all Algebra I row does not clear; none fails. The catalog lower-central rows on the page-read options rerun the same rule on transcription-1 options where stored (item bootstrap seed 11, 1,000 replicates; cluster bootstrap seed 20260916, 2,000 replicates); no new transcription was made for those cells and no second transcription verifies them.

Cross-family verification. A third transcription of the same 606 crops by a model of a different family was pre-registered after the repaired layer was scored and hashed before any LLaVA output was produced: bc891e7b5894216e662b1c78dae3dc627303509ae67aa17bab1d0e520a037ef0. Transcriber: LLaVA-NeXT Vicuna 13B (llava-hf/llava-v1.6-vicuna-13b-hf, revision 745cfbdb14dfeff9bcc6eb731937102e88ea3024), bf16, greedy, 640 new tokens, seed 20260916, the prompt and stored page crops of Appendix A.4, never asked to solve or to pick a letter. The LLaVA transcription is compared with transcription 1 only, never with transcription 2 or the text layer, by score_fidelity under the thresholds above; an item is *cross-family verified* when that call returns faithful, and every other item is not, including 13 items with no parseable LLaVA output, 5 of them two-page crops whose image tokens exceeded the model’s 4,096-token context. Counts: 160 cross-family verified and 446 not, of 606; against the Qwen–Qwen verdicts, 153 of the 160 are repaired-verified and 7 are repaired-unverified, and 278 repaired-verified items do not verify across families. Algebra I: 102 cross-family verified (97 also repaired-verified); Algebra II: 58 (56). With a withheld quantity, 101 and 57. No language model was re-scored: the rates in Table 1 and Section 4.2 join the saved repaired-layer predictions and rank-preserving isomorph letters to the cross-family id lists, with the bootstrap and bar above. On the 96 Algebra I and 55 Algebra II items verified by both checks the verdicts are the same: every Algebra I row clears, the Algebra II Qwen rows clear, and Phi-4 Algebra II ($21/55 = 0.382$ [0.236, 0.509]

against modal C 0.273) does not. The export records the verdict, the per-item scores, and the LLaVA output for every item.

Qwen 14B amendment on the text layer. The original isomorph control excluded Qwen 14B at $592/606 = 0.977$ against a 0.98 reproduction floor. The preregistration above re-runs the 14B original masked-stem letters on the census text layer at batch size 1, keeps the floor at 0.98, and scores both perturbed arms whatever the agreement, with the label attached. Agreement was again $592/606 = 0.977$, label *below floor*; the 14 disagreeing item ids are listed in the export. Table 15 reports the arms so a reader can see them. It is not a pass of the control. The repaired-layer control in Table 2 covers Qwen 14B without a floor because the original and perturbed letters come from one run.

Cause of the 14B floor miss. The saved masked-stem 14B letters were scored under torch 2.12.1; both re-scorings ran under torch 2.14.0 with the same model snapshot, prompt, mask, seed, bf16 dtype, scaled-dot-product attention, and batch size 1, on the same GPU. The two later loads agree with each other on all 606 letters and, on the 14 disagreeing ids, on every per-letter log-probability to the last bit, and three further bf16 repeats of those 14 items did not flip a letter, so the miss is a systematic shift between torch versions, not run-to-run kernel noise. On the later run 6 of the 14 disagreements are exact ties between the top two letters (margin 0 nats, resolved to the earlier option by the argmax; the saved run had 3 exact ties among the 14), the other 8 have later-run top-two margins between 0.25 and 0.75 nats, and on one Algebra II item the saved run had preferred A by 6.75 nats where the later runs prefer D; the median margin on the other 592 items is 5.0 nats on both runs. Casting the logits head to fp32 recovers 4 of the 14 saved letters; torch.use_deterministic_algorithms with bf16 attention recovers none. The arms stay labelled below floor and are not promoted; the isomorph preregistration is unchanged, and a margin-based floor that scores agreement only on items whose top-two margin is at least 0.05 nats on both runs would be a new pre-registered amendment, which we have not made.

A.7 Item-level demand coding

Two language-model coders, GLM 5.2 and Kimi K3, each coded the 358 Algebra I withheld-quantity items under one fixed rubric with three codes: execution (to pick the key you must operate on the specific given quantities: solve, evaluate, compute, factor with these coefficients, rewrite this expression, create an equation from these numbers, graph from this equation, find this value), recognition or interpretation (you can pick the key by recognizing or interpreting the form, meaning, or property named in the wording without computing on the given numbers), and mixed (needs both, or the key needs a small computation that any option-form reading

would also give). Each coder received only the item id, stem, and options, with the key, every prediction, and every score field stripped by a helper script, and neither read the cluster split or the other coder’s output. Each coder’s README records the rubric verbatim with its sha256 (ae93b7c651614dd622bba6ff4a9eddb43c69b4b9ff67407d245365c18b11446 for the three rubric lines joined by newlines; 434ed271810ece24b0aa050e9424f0caaf7e3398d5654de7318a558505b6ef18 for the same lines as a bulleted text file). Item text came from the print-fidelity transcription where it existed (352 and 351 items) and from the text layer otherwise; the repaired layer did not yet exist when the coders ran. Coders could mark an item unreadable; GLM did so on two items whose options refer to a list absent from the transcribed text.

Codes: GLM 212 execution, 116 recognition or interpretation, 28 mixed, 2 unreadable; Kimi 249, 94, 15, 0; the cluster split 164, 151, 43. Cohen’s κ (unweighted, three classes, unreadable dropped): GLM versus Kimi 0.576 ($n = 356$, observed agreement 0.789); GLM versus cluster 0.071 (0.461); Kimi versus cluster 0.230 (0.564). GLM against Kimi: 195 items both execution, 82 both recognition or interpretation, 4 both mixed; 32 items GLM recognition and Kimi execution, 8 the reverse. Against the cluster tag, 86 of the 151 recognition-tagged items are coded execution by GLM and 86 by Kimi (79 by both), and 41 (GLM) and 18 (Kimi) of the 164 execution-tagged items are coded recognition or interpretation. Subsets used in Table 3: both coders execution 195 items (142 repaired-verified), both recognition 82 (61), both coders and the tag execution 97 (79), both coders and the tag recognition 51 (35). Rows are recomputed from the saved repaired-layer scores with the bootstrap above; the script and the per-subset id lists are in the export.

B Audit trail: text-layer and print-faithful rows

These tables are the rows on which earlier revisions made their claims, kept so a reader can follow each row from the text layer, through the print-faithful filter, to the repaired layer of Section 4.2. Nothing in this appendix is claimed on its own.

Options only. On the 1,487 scoreable clean items, both local Qwen models score 0.301 versus chance 0.250 and always-B 0.275 (7B interval [0.277, 0.326]; 14B [0.276, 0.323]). Letter agreement between models is 0.457. A with-stem sanity check on 7B is 0.460 on 200 items. A proprietary assistant scored through an audited no-tool transcript scores 0.359 ($n = 1493$, interval [0.334, 0.384]). No $n \geq 10$ cell has a 7B lower interval above that cell’s modal-letter frequency. Algebra I is 0.309 [0.265, 0.355] versus modal B 0.272 (111 of 408). On the 1,026 items whose key is not the cell modal letter, 7B is 0.318 [0.288, 0.347] versus chance 0.250. On 14B, Algebra I is 0.321 [0.279, 0.363] versus modal 0.272: the lower interval exceeds the modal frequency. That options-only

Algebra I row is not a claimed witness; the claimed object is the masked-stem condition. Recency on the proprietary scorer minus 7B is largest before 2015 (+0.112, $n = 178$), then +0.024 in 2015–2019 ($n = 578$) and +0.069 in 2020 and later ($n = 712$).

Masked stem on the text layer. Of 1,487 scored items, 272 had nothing withheld; the primary population is the other 1,215. First-run pooled rates on all 1,487 (7B 0.405, 14B 0.425) mix those 272 in and are not the claim. Table 9 gives the named cells. Geometry does not beat its modal letter 0.289 on 7B (0.319, $n = 166$, interval [0.253, 0.392]) and does not clear chance on 14B (0.301 [0.235, 0.373]). EQAO grade 6 does not clear chance on 7B (0.400, $n = 40$, interval [0.225, 0.575] versus 0.255). TIMSS knowing on this population does not beat its modal letter 0.409 on 7B (0.500, $n = 22$, interval [0.318, 0.682]). The first-run TIMSS knowing rate 0.657 ($n = 35$) was carried by items where nothing was withheld: of 35 items, 18 are knowledge-answerable without any quantity, 12 are cue-readable from the leftover option set, and 5 are unclear. One-sided binomial p against chance on the text-layer withheld-quantity population: Algebra I 5.7×10^{-9} (Qwen 7B), 4.5×10^{-12} (Qwen 14B), and 9.4×10^{-13} (Phi-4); Algebra II 2.9×10^{-5} (7B), 9.0×10^{-5} (14B), and 4.2×10^{-4} (Phi-4); Phi-4 geometry 9.9×10^{-6} (67/166). Against the cell modal frequency: Algebra I 9.4×10^{-7} , 2.2×10^{-9} , and 5.6×10^{-10} ; Algebra II 0.00147, 0.00343, and 0.011; Phi-4 geometry 0.00104. Phi-4 text-layer rows: Algebra I 0.422 [0.374, 0.472]; Algebra II 0.347 [0.286, 0.407]; Algebra I residual with none of the four catalog cues 0.380 [0.307, 0.460] ($n = 163$); non-modal-key accuracy 0.387 ($n = 261$); numeric options 0.366 ($n = 71$); geometry 0.404 [0.331, 0.476] ($n = 166$) with non-modal-key split 0.229 ($n = 118$); pooled masked-stem 0.401 [0.374, 0.426] versus options-only 0.291 (McNemar 277/143). Qwen 14B on the 40-item human sheet is 0.425; on the 28 items whose census row matches the printed packet, Cohen’s κ with the rater is 0.212 (human 9/28, 14B 14/28).

Table 8. Lower-central on every cell with $n \geq 10$ (items the rule fires on), text layer. Geometry and EQAO grade 6 clear the per-cell bar here and again on the page-read options (Table 4). Algebra II clears the bar and is not claimed (lower end on chance). Holm on this 17-cell slice is exploratory. The other 88 of 105 a priori tests are omitted.

Authority	Claim	n	Pass	95% CI	Chance	Bar
NY Regents	geometry	279	0.323	[0.265, 0.380]	0.250	yes
EQAO	grade 6	40	0.500	[0.350, 0.650]	0.254	yes
NY Regents	Algebra II	372	0.298	[0.253, 0.344]	0.250	yes (not claimed)
NY Regents	Algebra I	458	0.260	[0.223, 0.301]	0.250	no
TIMSS	reasoning	57	0.368	[0.246, 0.491]	0.249	no
TIMSS	applying	50	0.240	[0.120, 0.360]	0.249	no
TIMSS	knowing	41	0.293	[0.146, 0.439]	0.248	no
TEKS	Algebra I	61	0.213	[0.115, 0.328]	0.250	no
TEKS	grade 6	52	0.327	[0.192, 0.462]	0.250	no
TEKS	grade 5	49	0.347	[0.224, 0.490]	0.250	no
TEKS	grade 8	45	0.267	[0.156, 0.400]	0.250	no
TEKS	grade 7	41	0.317	[0.171, 0.463]	0.250	no
TEKS	grade 3	36	0.278	[0.139, 0.444]	0.250	no
TEKS	grade 4	35	0.314	[0.171, 0.486]	0.250	no
EQAO	grade 9	38	0.211	[0.079, 0.342]	0.257	no
EQAO	grade 3	33	0.242	[0.091, 0.394]	0.253	no
NAPLAN	year 7	14	0.214	[0.000, 0.429]	0.250	no

Table 9. Partial-input language-model channel on named cells, Qwen and Mistral, withheld-quantity population, text layer. Chance is mean $1/k$. Modal is the cell’s modal key-letter frequency on that population. Bar: $n \geq 10$ and lower 95% CI above chance and above that modal frequency. Algebra I and Algebra II are claimed on the repaired layer (Table 1), not here. Geometry: Phi-4 meets the cell bar on the text layer but not the split on items whose key is not the cell’s most common letter and is not claimed. TEKS grade 5 and grade 8 clear on Qwen 7B only and are not claimed. Mistral-7B-Instruct-v0.3 does not clear.

Claim	n	Qwen 7B	Qwen 14B	Mistral 7B	Chance	Modal	Bar
Algebra I	358	0.388 [0.341, 0.439]	0.416 [0.363, 0.469]	0.302 [0.257, 0.349]	0.250	0.271	Qwen; not Mistral
Algebra II	248	0.367 [0.306, 0.423]	0.359 [0.298, 0.415]	0.262 [0.210, 0.319]	0.250	0.278	Qwen; not Mistral
geometry	166	0.319 [0.253, 0.392]	0.301 [0.235, 0.373]	0.295 [0.229, 0.373]	0.250	0.289	none
EQAO grade 6	40	0.400 [0.225, 0.575]	0.425 [0.300, 0.575]	0.150 [0.050, 0.275]	0.255	0.475	none
TIMSS knowing	22	0.500 [0.318, 0.682]	0.591 [0.409, 0.773]	0.364 [0.136, 0.545]	0.248	0.409	none
TIMSS applying	24	0.375 [0.208, 0.583]	0.375 [0.208, 0.583]	0.167 [0.042, 0.333]	0.248	0.333	none
TIMSS reasoning	42	0.286 [0.167, 0.429]	0.310 [0.167, 0.452]	0.238 [0.119, 0.381]	0.261	0.286	none
TEKS grade 5	38	0.474 [0.316, 0.632]	0.368 [0.211, 0.526]	0.184 [0.079, 0.316]	0.250	0.263	7B only; not claimed
TEKS grade 8	18	0.667 [0.444, 0.889]	0.611 [0.389, 0.833]	0.389 [0.167, 0.611]	0.250	0.389	7B only; not claimed

Table 10. Information ladder on text-layer withheld-quantity Algebra I ($n = 358$) and Algebra II ($n = 248$), Qwen 7B and 14B. Each rung adds one kind of information; the gap between masked-stem and with-stem is what the given quantities buy.

Claim	Model	Chance	Modal	Options	Masked	With stem
Algebra I	Qwen 7B	0.250	0.271 [0.226, 0.318]	0.313 [0.268, 0.360]	0.388 [0.341, 0.439]	0.508 [0.461, 0.559]
Algebra I	Qwen 14B	0.250	0.271 [0.226, 0.318]	0.332 [0.285, 0.380]	0.416 [0.363, 0.469]	0.575 [0.525, 0.626]
Algebra II	Qwen 7B	0.250	0.278 [0.222, 0.331]	0.274 [0.218, 0.335]	0.367 [0.306, 0.423]	0.403 [0.339, 0.464]
Algebra II	Qwen 14B	0.250	0.278 [0.222, 0.331]	0.286 [0.234, 0.343]	0.359 [0.298, 0.415]	0.440 [0.379, 0.500]

Table 11. With-stem letter-argmax ceiling versus masked-stem on text-layer withheld-quantity Algebra I ($n = 358$). Bar: lower 95% CI above chance 0.250 and modal B 0.271. Mistral’s ceiling is its masked-stem rate.

Scorer	With stem	Masked stem	Bar
Mistral 7B	0.304 [0.257, 0.352]	0.302 [0.257, 0.349]	chance only
Qwen 7B	0.508 [0.461, 0.559]	0.388 [0.341, 0.439]	both
Qwen 14B	0.575 [0.525, 0.626]	0.416 [0.363, 0.469]	both
Phi-4	0.497 [0.447, 0.553]	0.422 [0.374, 0.472]	both

Table 12. Every row claimed on the text layer, recomputed on print-faithful items and on the not-faithful complement (the filter step). Catalog rows: the lower-central rule; bar is item lower bound above chance. Language-model rows: masked-stem; bar is lower bound above chance 0.250 and above the modal key-letter frequency of that subset (Algebra I: 0.271 text layer, 0.255 faithful, 0.286 not faithful; Algebra II: 0.278, 0.296, 0.323). Algebra II cleared chance but not the modal-letter bar on its 115 faithful items and was dropped at this step; it re-enters on the repaired layer (Table 1). Geometry kept its rate and lost its interval at $n = 99$ and re-enters on the page-read options (Table 4).

Row	Scorer	Text layer			Print-faithful			Not faithful		
		n	Rate [95% CI]	Bar	n	Rate [95% CI]	Bar	n	Rate [95% CI]	Bar
Geometry lower-central	rule	279	0.323 [0.265, 0.380]	yes	99	0.323 [0.232, 0.414]	no	178	0.326 [0.253, 0.399]	yes
EQAO grade 6 lower-central	rule	40	0.500 [0.350, 0.650]	yes	14	0.571 [0.286, 0.857]	yes	24	0.458 [0.250, 0.667]	no
Algebra I masked-stem	Qwen 7B	358	0.388 [0.341, 0.439]	yes	184	0.467 [0.391, 0.543]	yes	168	0.292 [0.220, 0.363]	no
Algebra I masked-stem	Qwen 14B	358	0.416 [0.363, 0.469]	yes	184	0.429 [0.359, 0.500]	yes	168	0.393 [0.321, 0.464]	yes
Algebra I masked-stem	Phi-4	358	0.422 [0.374, 0.472]	yes	184	0.467 [0.397, 0.538]	yes	168	0.363 [0.292, 0.440]	yes
Algebra II masked-stem	Qwen 7B	248	0.367 [0.306, 0.423]	yes	115	0.348 [0.261, 0.435]	no	124	0.395 [0.306, 0.476]	no
Algebra II masked-stem	Qwen 14B	248	0.359 [0.298, 0.415]	yes	115	0.383 [0.287, 0.461]	no	124	0.339 [0.250, 0.419]	no
Algebra II masked-stem	Phi-4	248	0.347 [0.286, 0.407]	yes	115	0.348 [0.270, 0.435]	no	124	0.347 [0.266, 0.427]	no

Table 13. Masked-stem rows by the verb class of the publisher’s tag on the text layer and on print-faithful items (the filter step). Bar: $n \geq 10$, lower 95% CI above chance 0.250 and above the modal key-letter frequency of that subset. On print-faithful items the execution-tagged Algebra I class ($n = 75$) cleared on no model; on the repaired-verified population ($n = 130$) it clears on all three (Table 3).

Claim	Tag class	n	Qwen 7B	Qwen 14B	Phi-4	Modal	Bar
Algebra I	execution, text layer	164	0.293 [0.220, 0.360]	0.378 [0.311, 0.457]	0.366 [0.293, 0.439]	B 0.299	14B only
Algebra I	execution, print-faithful	75	0.360 [0.253, 0.480]	0.333 [0.240, 0.440]	0.387 [0.293, 0.507]	B 0.293	none
Algebra I	recognition or interpretation, print-faithful	83	0.518 [0.410, 0.614]	0.446 [0.349, 0.554]	0.506 [0.398, 0.614]	D 0.265	all three
Algebra I	mixed, print-faithful	26	0.615 [0.423, 0.808]	0.654 [0.462, 0.846]	0.577 [0.385, 0.769]	C 0.346	all three
Algebra II	execution, text layer	120	0.392 [0.308, 0.483]	0.400 [0.317, 0.475]	0.367 [0.283, 0.450]	C 0.300	Qwen only
Algebra II	execution, print-faithful	56	0.393 [0.250, 0.518]	0.411 [0.286, 0.536]	0.339 [0.214, 0.464]	C 0.286	none
Algebra II	recognition or interpretation, print-faithful	38	0.316 [0.184, 0.474]	0.342 [0.211, 0.500]	0.316 [0.184, 0.474]	B 0.395	none
Algebra II	mixed, print-faithful	21	0.286 [0.095, 0.476]	0.381 [0.190, 0.571]	0.429 [0.238, 0.619]	A 0.333	none

Table 14. Option-numeral isomorph control on print-faithful Algebra I masked-stem items, text layer (the filter step). Original is the re-scored rate from the same model load; it differs from the saved rates by at most one item per row on Qwen 7B (11 of 606 letters changed on re-scoring) and by none on Phi-4. Rank-preserving keeps the order of the options’ leading numbers (primary arm); scrambled does not. Paired Δ is rank-preserving minus original on items with at least one option numeral ($n = 136, 56, 66$). Bar: lower 95% CI above chance 0.250 and above the subset modal-letter frequency (0.255, 0.265, 0.293), for the rank-preserving and the scrambled arm. Qwen 14B was not scored here: its re-scored original letters agreed with the saved letters on $592/606 = 0.977$, below the pre-registered floor of 0.98 (Table 15). The repaired-layer control that covers all three scorers is Table 2.

Scorer	Items	n	Original	Rank-preserving	Scrambled	Paired Δ [95% CI]	Bar
Qwen 7B	faithful	184	0.473 [0.397, 0.549]	0.391 [0.321, 0.462]	0.391 [0.321, 0.462]	-0.110 [-0.199, -0.015]	yes / yes
Qwen 7B	recognition or interpretation	83	0.530 [0.422, 0.627]	0.361 [0.253, 0.458]	0.410 [0.301, 0.518]	-0.250 [-0.375, -0.125]	no / yes
Qwen 7B	execution	75	0.360 [0.253, 0.480]	0.387 [0.280, 0.507]	0.320 [0.213, 0.427]	+0.030 [-0.106, 0.152]	no / no
Phi-4	faithful	184	0.467 [0.397, 0.538]	0.408 [0.342, 0.478]	0.446 [0.375, 0.516]	-0.081 [-0.169, 0.015]	yes / yes
Phi-4	recognition or interpretation	83	0.506 [0.398, 0.614]	0.398 [0.289, 0.506]	0.410 [0.301, 0.518]	-0.161 [-0.304, -0.036]	yes / yes
Phi-4	execution	75	0.387 [0.293, 0.507]	0.373 [0.267, 0.480]	0.440 [0.333, 0.560]	-0.015 [-0.152, 0.121]	no / yes

Table 15. Qwen 14B option-numeral arms on the text layer under the amendment of Appendix A.6. Re-scored original letters agreed with the saved letters on $592/606 = 0.977$, below the 0.98 floor, so this table is labelled below floor and is not a pass of the control; the cause is in Appendix A.6. Paired Δ is rank-preserving minus original on items with an option numeral. Bar is for the rank-preserving arm against chance 0.250 and the subset modal-letter frequency.

Items	n	Original	Rank-preserving	Scrambled	Paired Δ [95% CI]	Bar
Algebra I, all	358	0.408 [0.355, 0.458]	0.360 [0.313, 0.413]	0.316 [0.271, 0.366]	-0.057 [-0.114, 0.000]	yes
Algebra I, faithful	184	0.418 [0.348, 0.489]	0.380 [0.315, 0.457]	0.370 [0.304, 0.446]	-0.051 [-0.132, 0.029]	yes
Algebra I, faithful, recognition-tagged	83	0.446 [0.337, 0.554]	0.386 [0.289, 0.494]	0.410 [0.301, 0.518]	-0.089 [-0.214, 0.054]	yes
Algebra I, faithful, execution-tagged	75	0.307 [0.213, 0.413]	0.333 [0.227, 0.440]	0.293 [0.187, 0.400]	+0.030 [-0.076, 0.136]	no
Algebra II, all	248	0.363 [0.302, 0.423]	0.323 [0.262, 0.379]	0.294 [0.238, 0.351]	-0.044 [-0.119, 0.026]	no
Algebra II, faithful	115	0.374 [0.287, 0.452]	0.339 [0.252, 0.435]	0.278 [0.200, 0.365]	-0.038 [-0.160, 0.085]	no